

UNIT IV

CHAPTER 4

Spread Spectrum Modulation

Syllabus

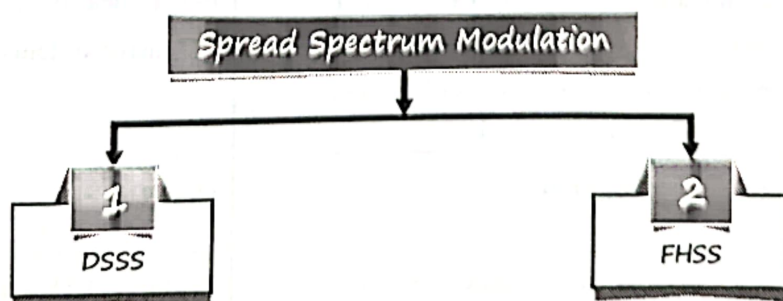
Use of Spread Spectrum , Direct Sequence (DS) Spread Spectrum, Spread Spectrum and Code Division Multiple Access (CDMA), Ranging Using DS Spread Spectrum , Frequency Hopping (FH) Spread Spectrum, Pseudorandom (PN) Sequences: Generation and Characteristics, Synchronization in Spread Spectrum Systems

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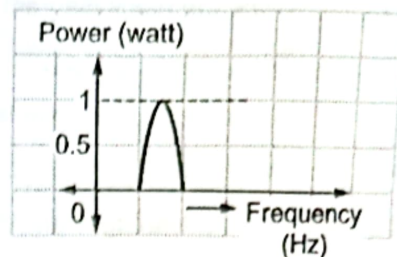


4.1 INTRODUCTION

A. Narrowband Signal

► What is narrowband signal? *(For reading purpose only...)*

The signal which utilizes narrower band of frequencies in the communication channels is known as the 'narrowband signal'. Typically narrowband signals are used for carrying signals like audio spectrums of limited bandwidth. It makes use of lesser bandwidth. Refer Fig. 4.1.1.

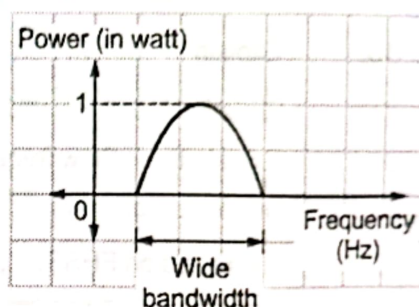


(1F1) Fig. 4.1.1 : Narrowband

B. Wideband Signal

► What is a wideband signal? *(For reading purpose only...)*

The signal which makes use of wide band of frequencies in the communication channels is known as 'wideband signal'. Refer Fig. 4.1.2.



(1F2) Fig. 4.1.2 : Wideband spectrum

C. Comparison between Narrowband and Wideband Signal *(For reading purpose only ...)*

Sr. no.	Parameter	Narrowband signal	Wideband signal
1	Definition	The signal which utilizes narrower band of frequencies in the communication channels is known as the narrowband signal.	The signal which makes use of wide band of frequencies in the communication channels is known as wideband signal.
2	Advantages	Better range of reception as design of narrower filters is simple. It can be used when slow data streams are to be transmitted	As wide bandwidth is used, faster communication achieved. It allows reduction of narrowband noise It provides high degree of security of data.
3	Disadvantages	Slow communication speed as narrower bandwidth is used. Noise can affect the signal easily as less security measures are used.	Better linearity of filters is needed. Expensive systems.
4	Modulation schemes used	AM, FM, SSB, BPSK, etc.	OFDM, GMSK, etc.
5	Applications	AM/FM radio, satellite downlinks, GPS signals, etc.	Wi-Fi, 4G LTE, OFDM, etc.

4.2 CONCEPT OF SPREADING THE SPECTRUM

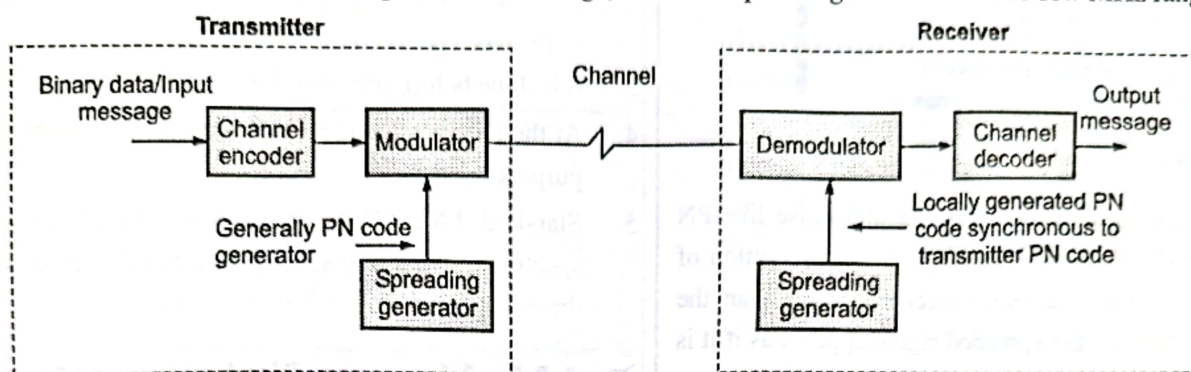
UQ. Draw the block diagram of spread spectrum digital communication and explain the various blocks.

SPPU - (E&TC-Sem. 5) Dec. 17, 8 Marks

✓ Ans. :

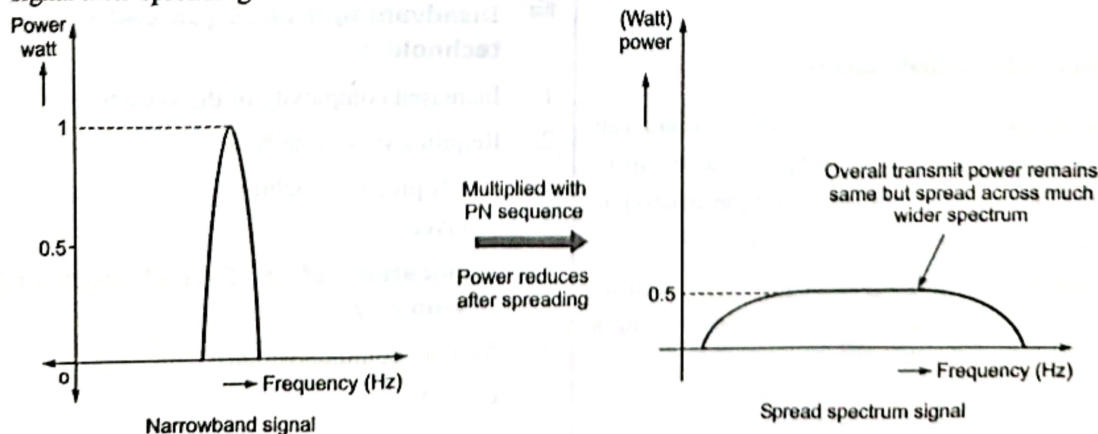
□ **Definition :** The process in which narrowband signal is converted to wideband signal is called as 'spreading of spectrum's.

Concept of spread spectrum: If the signal is in kHz range, then after spreading it is converted to few MHz range.



(1F3)Fig. 4.2.1 : Block diagram of spread spectrum system

1. Refer Fig. 4.2.1. It shows block diagram of spread spectrum system. Binary data is given as the input to the channel encoder. This binary data is narrowband signal. Generally channel encoder adds **redundant bits** to the input binary data bits and provides it to the modulator.
2. The modulator block actually comprises of multiplier and digital CW modulators. Multiplier has the other input from PN sequence or spreading code generator. This spreading code is wideband noise like signal. When the binary data is multiplied (**modulo - 2 added**) with the PN code, a wideband signal is obtained at the output.
3. This is **spreaded binary code**. This is then modulated using MPSK or MFSK modulators and then transmitted over channel. At the receiver, the signal is first demodulated by coherent or non coherent detection methods. Once the spreading code is detected back, it is again multiplied with the locally generated PN code. Then the original binary data is recovered back.
4. The power of the narrowband signal is reduced to almost half or even less than that, after spreading the signal. This is shown in Fig. 4.2.2. Refer Fig. 4.2.2. But the overall transmit power remains same even after spreading the signal. This power gets distributed over wider spectrum. In earlier digital communication, entire power was concentrated in tight bandwidth. This is not done in spread signal. For example, **6 MHz, 0.3 mW** signal may get converted into **22 MHz, 0.1mW** signal after spreading.



(1F4)Fig. 4.2.2 : Concept of spread spectrum

4.2.1 Need of Spreading the Spectrum

UQ. What is need of spread spectrum modulation technique?

SPPU - (E&TC-Sem. 5) May 16, 4 Marks

☒ **Ans. :**

1. Security
2. Resistance to jamming
3. Resistance to fading
4. Capability of multiple access

▶ 1. Security

In the spreading process, the sequential noise like PN (pseudo code) is used. Spreading allows encryption of the signal. To all the other receivers, other than the intended receiver, the spreaded signal appears as if it is noise. It has been used in military applications since 1940, as security is the major necessity. Later, it was used in commercial mobile applications.

▶ 2. Resistance to jamming

In spread spectrum signals, the jamming power is spread all over the spectrum. As the spectrum is wide enough, the overall effect of jamming is very much less than the background noise. On the contrary, in narrowband signals, as the bandwidth is less, the effect of jamming power is significant. Jamming severely degrades the quality of signal.

▶ 3. Resistance to fading

Wide bandwidth signal supports some sort of frequency diversity. Therefore, signal faces multipath fading very rarely.

▶ 4. Capability of multipath access

- With multiple access techniques, multiple users can transmit signals at a same time using the same carrier frequency. CDMA (code division multiple access) is implemented using spread spectrum technology.
- The SS (spread spectrum) signal is generated from an already data modulated carrier. This is done by using a wideband spreading signal.

4.2.2 Conditions for Spread Spectrum Signal

GQ. State conditions for spread spectrum signal.

☒ **Ans. :**

1. The transmission bandwidth should be much greater than the minimum signal bandwidth required for transmission.
2. For the purpose of spreading, codes should be used and these codes need to be independent of the data sequence.
3. It is done before transmission.
4. At the receiver, same code is necessary for despreading purpose.
5. Standard FM, PCM, etc. are not called as spread spectrum modulation schemes, as they do not satisfy all these conditions.

4.2.3 Advantages, Disadvantages and Applications of Spread Spectrum Technology

GQ. State advantages, disadvantages and applications of spread spectrum technology.

☒ **Ans.:**

☞ Advantages of SS (Spread Spectrum) technology

1. It has relatively high security.
2. SS system has better noise immunity by resistance to narrowband jamming of the signal.
3. This system supports better performance in multipath fading environment.
4. SS supports multiple access schemes like CDMA.

☞ Disadvantages of SS (Spread Spectrum) technology

1. Increased complexity of the system.
2. Requires very large bandwidth.
3. Needs precise synchronization between transmitter and receiver.

☞ Applications of SS (Spread Spectrum) technology

1. Military communication
2. CDMA technology
3. Mobile communication

4.2.4 Classification of Spread Spectrum Modulation Techniques

There are four types of spread spectrum techniques depending on the type of insertion of PN code

1. DSSS : Direct Sequence Spread Spectrum
2. FHSS : Frequency Hopped Spread Spectrum
3. THSS : Time Hopping Spread Spectrum
4. Hybrid Spread Spectrum

4.3 PSEUDO NOISE SEQUENCE (PN) CODE

GQ. Explain PN sequences generator with neat block diagram.

UQ. What is PN sequence ?

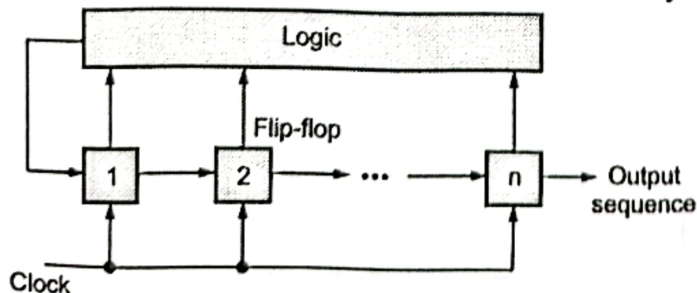
SPPU - (E&TC-Sem. 5) May 15, Dec. 16,

May 17, 2 Marks

✓ **Ans. :**

□ **Definition :** PN code is the deterministic and periodic binary sequence generated by feedback logic circuit. It provides noise like waveform at the output.

1. Refer Fig. 4.3.1. It shows the general block diagram of PN sequence generator. The design of PN sequence generator (**feedback shift register**) consists of 'n' number of flip-flops and the feedback logic circuit. It forms the multiloop feedback circuit. A single timing clock is given to the generator.
2. It is the shift register. When the clock pulse is applied, the output of the flip-flop is shifted to the input of the next flip-flop and next state is obtained. The output of PN sequence is taken at the final stage flip-flop.
3. At every state, the feedback logic also contributes in determination of the state as it is in feedback circuitry.



(1F5)Fig. 4.3.1: PN sequence generator

4. Let $m_j(k)$ denote the state of the flip-flop at k^{th} clock pulse. The state can either be '0' or '1'. Hence, the state

of the shift register after k^{th} clock pulse can be given as a set of $\{m_1(k), m_2(k), m_3(k), \dots, m_n(k)\}$;

where, $k \geq 0$

5. Let for initial state, $k = 0$, Therefore,

$$m_j(k+1) = m_{j-1}(k), \quad \{k \geq 0 ;$$

$$\{1 \leq j \leq n$$

6. If 'n' number of the flip-flop are used, then the number of states in PN sequence is equal to 2^n . If the feedback logic consists of modulo-2 adders, then the '0' state, i.e., output state of all the flip-flops is zero, is not allowed.
7. This is because, if the initial state is kept zero and as there are modulo-2 adders in the feedback logic, then the shift register will continue with the same state and will not advance to next state. This is deadlock situation. Hence, the number of states in the PN generator = 2^{n-1} . (One state is omitted which is all zero state). Maximal length sequence or m sequence: The PN sequence is known as maximal length sequence when the states are equal to 2^{n-1} .

4.3.1 Comparison between Pseudo Random Signal and the Random Signal

GQ. What is the difference between the Pseudo random signal and the Random signal?

✓ **Ans. :**

Table 4.3.1 : Comparison between Pseudo random signal and the random signal

Sr. No.	Parameters	Pseudo random signal	Random signal
1.	Prediction	It is deterministic and periodic sequence; hence, can be predicted exactly.	It cannot be predicted exactly.
2.	Analysis	It can be analysed mathematically.	It needs statistical analysis.
3.	Nature of signal	It is known to both transmitter and receiver in spread spectrum systems.	It is totally random, so cannot be generated.

4.3.2 PN sequence as Pseudo Random

Why PN sequence is known as pseudo random noise like signal?

Q. Why PN sequence is known as pseudo random noise like signal?

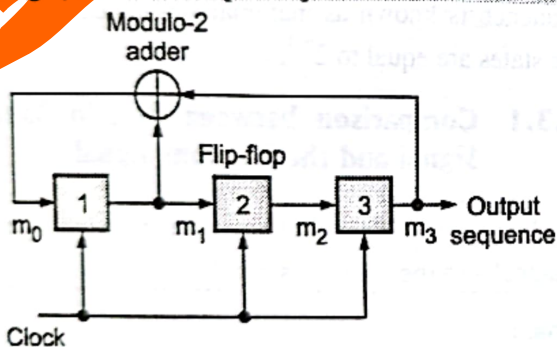
Ans. :

Even though the signal is deterministic, it appears as if it is white noise or a truly random signal to the unauthorized receiver. Hence, it is called as noise like signal.

4.3.3 Examples on PN Sequence

UEx. 4.3.1 SPPU - (E & TC-Sem. 5) May 17, 6 Marks

Design a three stage feedback shift register with proper taps to generate $N = 7$ PN sequence. Draw the generator block and if the initial state of shift register is 100 (from left to right). Find the output sequence.



(1F6) Fig. P. 4.3.1 : PN sequence generator with $n=3$

Soln. :

- Number of flip-flop = 3
Therefore, $n = 3$,
Hence, number of states $N = 2^n - 1 = 2^3 - 1 = 8 - 1 = 7$
- The modulo-2 adder can be simple EX-OR logic gate.

Truth table of EX-OR gate

Input		Output (Y)
A	B	
0	0	0
0	1	1
1	0	1
1	1	0

- Now looking at the diagram, it is clear that output of modulo-2 adder = $m_1 \oplus m_2$. Hence, the truth table considering state 100 as initial state is shown in Table P. 4.3.1.

Table P. 4.3.1 : Truth table of PN sequence generator

Clock	m_1	m_2	m_3
1	1	0	0
2	1	1	0
3	1	1	1
4	0	1	1
5	1	0	1
6	0	1	0
7	0	0	1
8	1	0	0

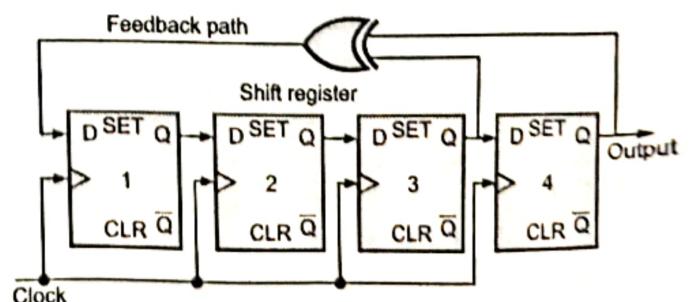
- The output sequence (output of the last stage flip-flop of the shift register is therefore, 0011101.... The sequence will repeat after 7th state as shown in Table P. 4.3.1.

Note : Any of the six states can be taken as initial state.

UEx. 4.3.2 SPPU - (E & TC-Sem. 5) May 16, 9 Marks

Generate the PN sequence for transmitting message through FHSS system. The period of PN sequence is $2^4 - 1 = 15$. The initial content of shift register is assumed to be 1100. Draw PN sequence generator.

Soln. :



(1F23) Fig. P. 4.3.2 : PN sequence generator

Clock pulse	Q ₄	Q ₃	Q ₂	Q ₁	Q ₃ ⊕ Q ₄	PN Code
1	1	1	0	0	0	1
2	1	0	0	0	1	1
3	0	0	0	1	0	0
4	0	0	1	0	0	0
5	0	1	0	0	1	0
6	1	0	0	1	1	1
7	0	0	1	1	0	0
8	0	1	1	0	1	0
9	1	1	0	1	0	1
10	1	0	1	0	1	1
11	0	1	0	1	1	0
12	1	0	1	1	1	1
13	0	1	1	1	1	0
14	1	1	1	1	0	1
15	1	1	1	0	0	1
16	1	1	0	0	0	1

Initial state

(1F24)Fig. P. 4.3.2(a)

4.3.4 Properties of Maximal Length Sequence / PN Sequence

UQ. What are the properties of maximal length sequences? Give the graphical representation of auto correlation property of random data and a PN sequence and explain.

SPPU - (E&TC-Sem. 5) Dec. 14, 8 Marks

UQ. State the properties of PN sequence with the help of 4 stage shift register.

SPPU - (E&TC-Sem. 5) May 15, 6 Marks.

Dec. 17, 6 Marks

UQ. Verify the three properties of PN sequence with the help of shift register.

SPPU - (E&TC-Sem. 5) Dec. 16, May 17, 6 Marks

✓ **Ans. :**

1. Balance property
2. Run property
3. Correlation property

1. Balance property

In each period of the PN sequence, the number of 1's is always one more than number of 0's. E.g. in the sequence in Ex. 4.3.1 the number of 1's are 4 and number of 0's are 3. Hence, number of 1's is one more than number of 0's.

2. Run length property

Definition of Run : A run is defined as the string of continuous identical binary digits.

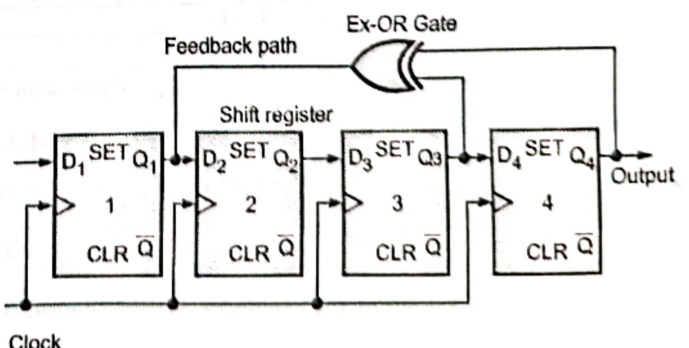
- New binary digit in the sequence changes the run. The run length is the number of binary digits in the run. The run property states that one half the runs of each type of length is equal to 1, about one fourth are of length 2, one eighth are of length 3 and so on.
- For a PN sequence generated by 'n' number of flip-flops, the total number of runs is $(N + 1)/2$, where $N = 2^n - 1$. Consider a PN code of length = 15. It means 4 flip-flops are used and 15 states are generated. This is 4 bit PN sequence generator as shown in Fig. 4.3.2.

4 bit PN sequence generator

UQ. Draw and explain 4 bit PN sequence generator and find maximum length sequence.

SPPU - (E&TC-Sem. 5) Dec. 15, 8 Marks

✓ **Ans. :**



(1F7)Fig. 4.3.2 : 4 bit PN sequence generator

Table 4.3.2 : The states obtained

Clock	Q ₁	Q ₂	Q ₃	Q ₄	PN code
1	1	0	0	0	0
2	0	1	0	0	0
3	0	0	1	0	0
4	1	0	0	1	1
5	1	1	0	0	0
6	0	1	1	0	0
7	1	0	1	1	1
8	0	1	0	1	1
9	1	0	1	0	0
10	1	1	0	1	1
11	1	1	1	0	0
12	1	1	1	1	1
13	0	1	1	1	1
14	0	0	1	1	1
15	0	0	0	1	1
16	1	0	0	0	0

- The PN code obtained is 0001001101011110. In this PN code sequence,

- Total number of runs = 8
- One run of length = 3 = $1/8^{\text{th}}$ of runs of length 3
- Two runs of length = 2 = $1/4^{\text{th}}$ of runs of length 2
- Four runs of length = 1 = $1/2$ runs of length 1

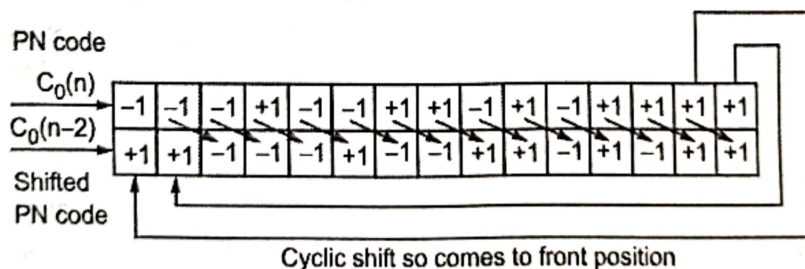
If the sequence has more number of states, then it goes on. This is known as 'the run length property of the PN sequence'.

3. Cyclic shift property / Autocorrelation Property

Consider a PN sequence shown in Fig. 4.3.2. The PN sequence is 000100110101111. We will allocate -1 for logic '0' and +1 for logic '1'. Hence, the sequence becomes,

PN code sequence →	0	0	0	1	0	0	1	1	0	1	0	1	1	1	1
+1 and -1 level for PN codes →	-1	-1	-1	+1	-1	-1	+1	+1	-1	+1	-1	+1	+1	+1	+1

- This code sequence is cyclically shifted in time, For example, the code shown above is shifted by two time periods and last two bits are allocated to first positions, as it is cyclic shift. Hence, we get,



(1F26) Fig. 4.3.3 : Cyclic shift

- Determining the correlation of these two sequences we get,

$$= \frac{1}{N} \sum C_0(n) \cdot C_0(n-2) \dots$$

Here $N = 15$ = length of PN sequence. Determination of correlation is done by taking product of $C_0(n)$ and $C_0(n-2)$.

Hence we get,

PN code →	$C_0(n)$	-1	-1	-1	+1	-1	-1	+1	+1	-1	+1	-1	+1	+1	+1
Shifted PN codes →	$C_0(n-2)$	+1	+1	-1	-1	-1	+1	-1	-1	+1	+1	-1	+1	-1	+1
Product	$C_0(n) C_0(n-2)$	-1	-1	+1	-1	+1	-1	-1	-1	-1	+1	+1	-1	+1	+1

$$\therefore \sum C_0(n) C_0(n-2) = -1$$

$$\therefore \text{Correlation } R(\tau) = \frac{1}{N} \sum C_0(n) C_0(n-2)$$

$$\therefore R(\tau) = \frac{1}{15} \sum C_0(n) C_0(n-2) = \frac{-1}{15}$$

- This is true for any shift. Hence it can be concluded that, the correlation for any shift other than zero is $-\frac{1}{N}$. For shift 0,

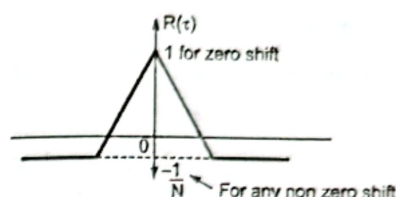
$$\text{Correlation} = \frac{1}{N} \sum C_0(n) C_0(n)$$

$$\therefore R(\tau) = \frac{1}{N} \sum C_0^2(n)$$

$$\therefore R(\tau) = \frac{1}{N} \sum 1 = \frac{N}{N}$$

$$\therefore R(\tau) = 1$$

$$\text{Correlation of PN sequence} = \begin{cases} 1 & \text{if shift} = 0 \\ -\frac{1}{N} & \text{For any, other non-zero shift} \end{cases}$$



(118) Fig. 4.3.4 : Correlation diagram for PN sequence

4.4 A NOTION OF SPREAD SPECTRUM

(For reading purpose only..)

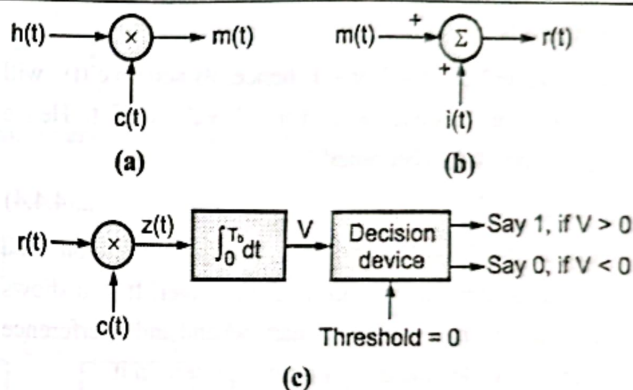
- The spread spectrum signals provide better protection against the **narrowband jamming signals**. These jamming signals may consist of fairly powerful broadband noise or multitone waveform which may disrupt the communication signals.
- It is provided by making the signal occupy wider bandwidth than **minimum transmission bandwidth** required. Signal is transmitted by hiding behind the wide bandwidth signal. It also gives the spreading signal an appearance of noise like signal. Therefore, it can smoothly pass through the channel without anyone listening to the signal. Hence, better security is provided to the signal.

4.4.1 Method of Spreading the Signal

GQ. Explain ideal system of spread spectrum technology.

Ans. : Refer Fig. 4.4.1.

- Binary data sequence (user data) used is always in NRZ (Non Return to Zero) format. Let $b(t)$ be the binary data sequence with levels ± 1 and $c(t)$ be the PN code.
- $b(t)$ is the narrowband signal and $c(t)$ is wideband signal. $b(t)$ and $c(t)$ are applied to the product modulator or the multiplier.



(119) Fig. 4.4.1 : Ideal system of spread spectrum technology (a) transmitter (b) channel (c) receiver

- As per Fourier transform theory, the multiplication of the narrowband and wideband signals will give convolution of the spectrum of two individual signals. Hence, the product of the narrowband spectrum and the wideband spectrum is nearly equal to the wideband spectrum. Hence, it can be said as PN code spreads the signal.
- After multiplication process, the user data or the information signal $b(t)$ is actually chopped into number of small times increments as shown in Fig. 4.4.1(b).
- These small increments are known as chips. Hence,

$$x(t) = b(t) c(t) \quad \dots(4.4.1)$$
- At the receiver, signal $r(t) = [x(t) + i(t)]$ is received. where, $i(t)$ = interference added
Therefore, $r(t) = b(t) c(t) + i(t) \quad \dots(4.4.2)$

7. To detect the original information signal back at the receiver, the received signal is given to demodulator circuit.
8. Refer Fig. 4.4.1(c). Demodulator consists of multiplier, integrator and the decision device as shown in Fig. 4.4.1(c). Locally generated PN code is provided to the multiplier. For generation of the exact PN code as that of transmitter, the transmitter and receiver has to maintain synchronism.
9. Hence the multiplier output is given as,

$$m(t) = c(t) r(t)$$

Putting Equation (4.4.2) in above equation we get,

$$m(t) = c^2(t) b(t) + c(t) i(t) \quad \dots(4.4.3)$$

11. From the Equation (4.4.3), it is clear that information signal is multiplied by square of the wideband PN code and unwanted interference signal is multiplied by single PN signal.

$c(t)$ can either be +1 or -1, hence, its square $c^2(t)$ will always be positive +1, for all values of t . Hence Equation (4.4.3) becomes

$$m(t) = b(t) + c(t) i(t) \quad \dots(4.4.4)$$

Equation (4.4.4) shows that binary information data signal is reproduced back at the receiver. It also shows that data signal received is narrowband and interference signal is wideband as $c(t)$ term is present in it.

13. Hence, this signal is now applied to LPF (Low Pass Filter). LPF will pass only narrowband signal and wideband interference is automatically filtered out.
14. This filtering action is performed by the integrator. Integration is carried out for one bit duration. Integrator's output is given to the decision device. If the incoming signal sample value is greater than the threshold set at the decision device, then logic '1' is detected otherwise logic '0'.

4.5 DIRECT SEQUENCE SPREAD SPECTRUM(DSSS)

- UQ. Draw block diagram of DS-PSK transmitter.

SPPU - (E&TC-Sem. 5) Dec. 14, 2 Marks

- UQ. Draw the block diagram of DSSS system transmitter and receiver. Write functional names inside the block and input signal for each block and explain.

SPPU - (E&TC-Sem. 5) May 17, 6 Marks

- UQ. Explain DS-BPSK transmitter and receiver with suitable block diagram.

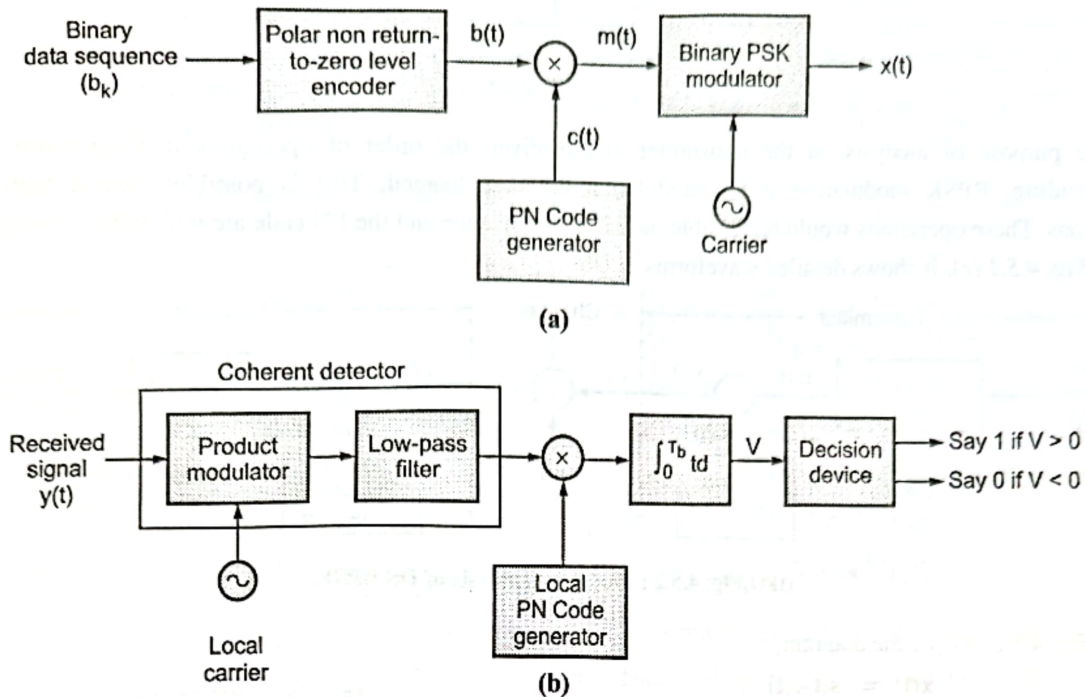
SPPU - (E&TC-Sem. 5) May 17, 3 Marks

- UQ. Explain DSSS transmitter and receiver with neat waveforms.

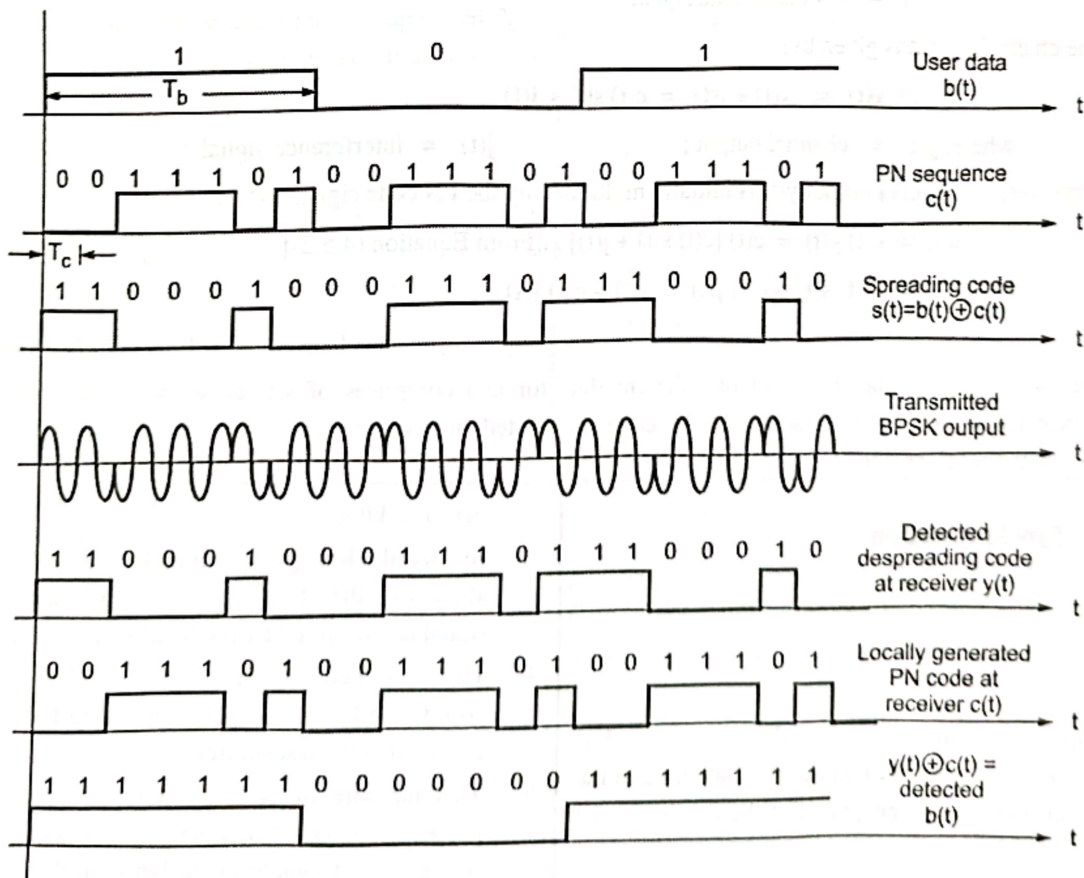
SPPU - (E&TC-Sem. 5) May 18, 8 Marks

✓ Ans. :

- Refer Fig. 4.5.1 (a). In DSSS, signal undergoes two stages of modulation. Initially binary data input b_k is converted into **bipolar NRZ format**. This NRZ signal is then given to the mixer. As this is the input data sequence, it is narrowband signal.
- The other input to the mixer is the wideband PN code $c(t)$. Modulo-2 addition (**EX-ORing operation**) is carried out and output is provided to the next stage which is **PSK modulator**. This output of the multiplier is known as 'spreading code'. It is now the wideband signal whose bandwidth is almost equal to the PN code signal.
- Therefore, the transmitted signal is DS/BPSK signal. In BPSK, two phases, 0 and π are present depending on the polarity of the data sequence. The waveforms are as shown in Fig. 4.5.1(c).
- Refer Fig. 4.5.1 (b). At the receiver, the received signal is first given to the product modulator and low pass filter combination. The **cut-off frequency** of the low pass filter is equal to the original message signal. Coherent detection is carried out for detection of the spreading code from the BPSK output. The output of the coherent detector is the spreading code.
- This spreading code and locally generated PN code are then multiplied. Digitally, modulo-2 addition or EX-OR operation is carried out. The condition of this PN code generation is that, it has to be same as that of the PN code used at the transmitter. This process is called as 'de-spreading'.
- After the de-spreading of the signal, it is given to the integrator. Integration is carried out for one bit duration.
- Integrator's output is provided to the decision device. The threshold is set for logic '1' and logic '0'. If the incoming bit is above the threshold, it is declared as logic '1' otherwise logic '0'. If the bit is exactly equal to the threshold, then the decision device takes the decision randomly.



(1F10) Fig. 4.5.1 : (a) DSSS transmitter and (b) DSSS receiver



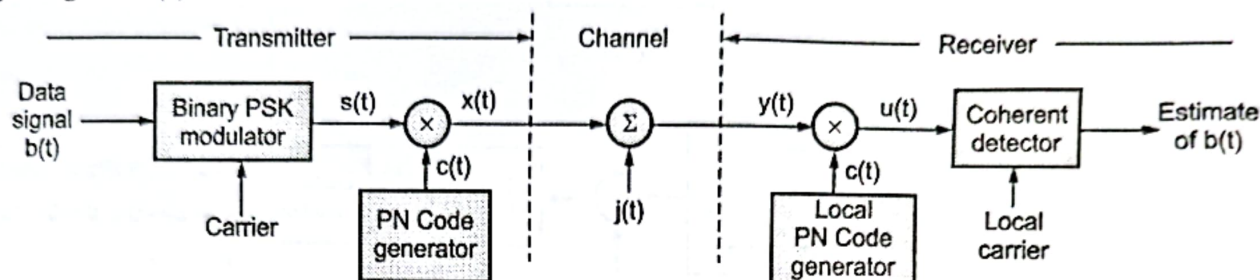
(1F11) Fig. 4.5.1(c) : DS / BPSK waveforms

4.5.1 Model for Analysis for DS / BPSK

GQ. Describe the model used for analysis of DS/ BPSK.

Ans. :

- For the purpose of analysis, at the transmitter and receiver, the order of operations like **spectrum spreading**, **de-spreading**, BPSK modulation and demodulation are interchanged. This is possible because both are linear operations. These operations would be feasible only if data sequence and the PN code are maintaining the synchronism. Refer Fig. 4.5.1 (c). It shows detailed waveforms of DS/BPSK.



(1F12)Fig. 4.5.2 : Model for analysis of DS BPSK

- Refer Fig. 4.5.2. As per the diagram,

$$x(t) = s(t)c(t)$$

...(4.5.1)

where, $x(t)$ = spreading signal ;

$s(t)$ = BPSK output signal

$c(t)$ = PN coded data signal

Now, the channel output is given by,

$$y(t) = x(t) + j(t) = c(t) s(t) + j(t)$$

...(4.5.2)

where, $y(t)$ = channel output ;

$j(t)$ = interference signal

- At the receiver, the channel output $y(t)$ is initially multiplied by the PN code signal $c(t)$. Hence,

$$u(t) = c(t) y(t) = c(t) [c(t) s(t) + j(t)] \text{ ...[From Equation (4.5.2)]}$$

$$= c^2(t) s(t) + c(t) j(t) = s(t) + c(t) j(t)$$

...(4.5.3)

[since $c^2(t) = 1$, as per PN code properties]

- Equation (4.5.3) shows that the input of coherent detector $u(t)$ comprises of $s(t)$ added with the code modulated interference signal $c(t) \cdot j(t)$. Presence of the code modulated interference part indicates requirement of increased reliability at the receiver side.

4.5.2 Synchronization

GQ. Explain the concept of synchronization used in spread spectrum technology.

Ans. :

- For proper **detection** of the original message signal at the receiver, it is necessary that the **PN code** used at the transmitter and the one generated at the receiver must be in synchronism with each other.
- It can be provided by :**
 - Acquisition

(ii) Tracking

- In acquisition phase**, initially two PN codes are aligned to the short duration chip. Then the received signal is multiplied by the locally generated PN code.
- The multiplied sequence is then used to determine the correlation between locally generated PN code and the one used at the transmitter.
- This measure of correlation is then processed using proper decision rule and search strategy. This will give the degree of synchronism between them. And it will

also provide the solution if the PN codes are not in synchronism.

6. Like the carrier reference is generated at the coherent detector, in a similar way, using the **phase lock techniques tracking** is implemented.

4.5.3 Power Spectral Density of DS/BPSK Signal

GQ. Derive and draw power spectral density of DS/BPSK signal.

UQ. Derive the power spectral density of DS/BPSK signal.

SPPU - (E&TC-Sem. 5) May 17, 3 Marks

Hint : $S_x(f) \approx \frac{A^2 T_c}{4} \left[\left(\frac{\sin \pi (f - f_c) T_c}{\pi (f - f_c) T_c} \right)^2 + \left(\frac{\sin \pi (f + f_c) T_c}{\pi (f + f_c) T_c} \right)^2 \right]$

✓ Ans. :

Power spectral density of DSSS signal

- The power spectral density of DSSS signal depends on the modulation scheme and the pulse shape used. PSD of a random process is Fourier transform of autocorrelation function of that process.
- PSD of PAM pulse signal is given by,

$$S_x(f) = \frac{|P(f)|^2}{T_s} \sum_{k=-\infty}^{\infty} R(k) e^{-j2\pi f k T_s} \quad \dots(4.5.4)$$

where, $P(f)$ = Fourier transform of pulse;

$R(k) = a_i a_{i+k}$ = autocorrelation of data sequence

T_s = symbol duration

- If the data is uncorrelated, then

$$R(k) = \begin{cases} \overline{a_i^2} & K = 0 \\ \overline{a_i a_{i+1}} & K \neq 0 \end{cases}$$

$$\therefore R(k) = \begin{cases} \sigma_a^2 + m_a^2 & K = 0 \\ m_a^2 & K \neq 0 \end{cases} \quad \dots(4.5.5)$$

where, σ_a = Variance and m_a

m_a = mean of data amplitude sequence

- Putting value of $R(k)$ in Equation (4.5.4) we get,

$$\begin{aligned} S_x(f) &= \frac{|P(f)|^2}{T_s} \sum_{k=-\infty}^{\infty} R(k) e^{-j2\pi f k T_s} \\ \therefore S_x(f) &= \frac{|P(f)|^2}{T_s} \left(\sigma_a^2 + m_a^2 \sum_{k=-\infty}^{\infty} e^{-j2\pi f k T_s} \right) \\ \therefore S_x(f) &= \frac{|P(f)|^2}{T_s} \left(\sigma_a^2 + m_a^2 \sum_{k=-\infty}^{\infty} \delta\left(f - \frac{k}{T_s}\right) \right) \\ &= \frac{\sigma_a^2}{T_s} |P(f)|^2 + \frac{m_a^2}{T_s} \sum_{k=-\infty}^{\infty} \left| P\left(\frac{k}{T_s}\right) \right|^2 \delta\left(f - \frac{k}{T_s}\right) \quad \dots(4.5.6) \end{aligned}$$

- Fourier transform of square pulse is given as,

$$P(f) = T_s \text{sinc}(T_s f)$$

Putting this $P(f)$ in Equation (4.5.6) and also putting $m_a = 0$ and $\sigma_a^2 = 1$ for phase modulation, we get,

$$\therefore S_x(f) = T_s \text{sinc}^2(T_s f) \quad \dots(4.5.7)$$

Transmitted signal $\tilde{s}(t)$ is random process.

- Data and spreading code are independent sequences. Hence, autocorrelation function is the multiplication of the autocorrelation functions of two signals.

$$\therefore \tilde{s}(t) = \sqrt{P} \cdot a(t) \cdot b(t)$$

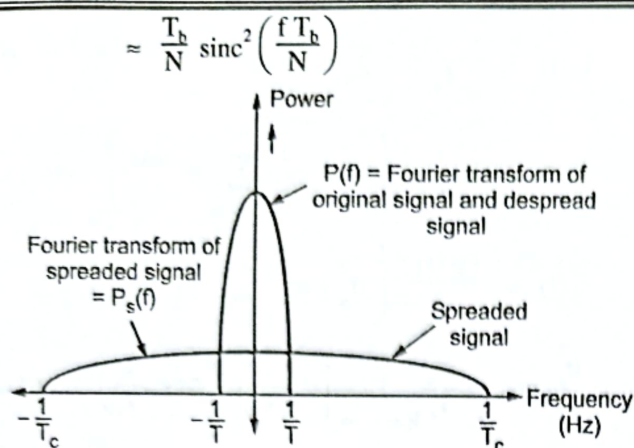
$$\begin{aligned} \therefore R_s(T) &= E \{ \tilde{s}(t) \tilde{s}^*(t+T) \} \\ &= E \{ a(t) b(t) a(t+T) b(t+T) \} \\ &= E \{ a(t) a(t+T) \} E \{ b(t) b(t+T) \} \\ &= R_a(T) R_b(T) \quad \dots(4.5.8) \end{aligned}$$

- Determining the power spectral density of $R_s(T)$ signal we get,

$$\begin{aligned} \therefore S(f) &= \int_{-\infty}^{\infty} S_b(\phi) S_a(f - \phi) d\phi \\ &= \int_{-\infty}^{\infty} T_b \text{sinc}^2(\phi T_b) T_c \text{sinc}^2\left([f - \phi] T_c\right) d\phi \\ &= \int_{-\infty}^{\infty} T_b \text{sinc}^2(\phi T_b) \frac{T_b}{N} \text{sinc}^2\left([f - \phi] \frac{T_b}{N}\right) d\phi \\ &\quad \dots \left[\text{since } T_c = \frac{T_b}{N} \right] \end{aligned}$$

when $N \gg 1$,

$$\therefore S_x(f) \approx \frac{T_b}{N} \text{sinc}^2\left(\frac{f T_b}{N}\right) \int_{-\infty}^{\infty} T_b \text{sinc}^2(\phi T_b) d\phi$$



(1F13) Fig. 4.5.3 : PSD of DS/BPSK signal

Observations from Fig. 4.5.3

- $S_b(t)$ is strong narrowband signal spectrum.
- Bandwidth of spread spectrum signal is N times the original information signal. Hence N is bandwidth expansion factor.

PSD of bandpass signal is,

$$S_s(f) \approx \frac{A^2 T_c}{4} \left[\left(\frac{\sin \pi (f - f_c) T_c}{\pi (f - f_c) T_c} \right)^2 + \left(\frac{\sin \pi (f + f_c) T_c}{\pi (f + f_c) T_c} \right)^2 \right]$$

Where A = signal amplitude

4.5.4 Advantages, Disadvantages and Applications of Direct Sequence Spread Spectrum (DSSS)

GQ. State advantages and applications of DSSS.

UQ. What are disadvantages of DSSS?

SPPU - (E&TC-Sem. 5) Dec. 17, 2 Marks

☒ **Ans. :**

Advantages of Direct Sequence Spread Spectrum (DSSS)

- It provides good discrimination against multipath signals.
- Performance of DSSS systems in the presence of noise is much better as compared to FHSS systems.
- It reduces the narrowband interference like jamming signals effectively.

Disadvantages of Direct Sequence Spread Spectrum (DSSS)

- System has long acquisition time.

- It requires wideband channel.
- High PN code data rates are needed.

Applications of Direct Sequence Spread Spectrum (DSSS)

- Military applications
- CDMA technology

4.6 SPREAD SPECTRUM AND CODE DIVISION MULTIPLE ACCESS(CDMA)

- In CDMA, each user is allocated with one code channels and these codes are orthogonally generated using spread spectrum technology.
- The advantage of CDMA system is that collisions are not destructive which means that the signals involved in a collision would be received with only a slight increase in error rate.
- Let each of the ' k ' users are transmitting data at same carrier ' f_0 ' using DSSS with code $g_i(t)$. Then each user is presented with same input waveform,

$$v(t) = \sum_{i=1}^k \sqrt{2P_s} g_i(t) d_i(t) \cos(\omega_0 t + \theta_i)$$

where, P_s = signal power

$g_i(t)$ = PRBS sequence

f_c = chip rate

$d_i(t)$ = data by ' i^{th} ' user

$\theta_i = i^{\text{th}}$ phase

- At the receiver k correlators are needed for k users at receiver 1, $u(t)$ is multiplied by $g_1(t)$ and also by $\sqrt{2} \cos(\omega_0 t + \theta_1)$ and (v_{01}) signal is obtained. It is then passed through integrate and dump filter and we get,

$$\begin{aligned} V_{01}' &= \sum_{i=1}^k \sqrt{P_s} g_1(t) g_i(t) d_i(t) \cos(\theta_i - \theta_1) \\ &= \sqrt{P_s} d_1(t) + \sum_{i=2}^k \sqrt{P_s} g_1(t) g_i(t) d_i(t) \cos(\theta_i - \theta_1) \end{aligned}$$

The product $g_1(t) \cdot g_i(t) = g_{1i}(t)$ and also replacing $\cos(\theta_i - \theta_1)$ by $\cos \theta_{1i}$ we get,

$$V'_{01} = \sqrt{P_s} d_1(t) + \sum_{i=2}^k \sqrt{P_s} g_{1i}(t) \cos \theta_{1i}$$

- Now there are $(k-1)$ independent interferers in (V'_{01}) signal. Hence total PSD is given as,

$$S_J(f) \approx (k-1) \frac{P_s}{4 f_b} \quad |f| \leq f_b$$

- The probability of a bit in error is

$$P_e = \frac{1}{2} \operatorname{erfc} \sqrt{2 \left[\frac{1}{k-1} \right] \left[\frac{f_c}{f_b} \right]}$$

- In slotted CDMA system, to ensure low error probability a gain of f_c / f_b must be adjusted such that

$$\frac{f_c}{f_b} \gg \frac{(k-1)}{2}$$

- In this, each transmitted signal is assumed to be of ' P_s ' power but practically it may not be true.
- Near-far problem :** When an unwanted user's receiver power is much larger than the received power of the desired user errors may occur. It is known as near-far problem. It limits the usage of DSSS systems to same received power applications. This problem can be reduced by adaptive power control.

4.7 RANGING USING DS SPREAD SPECTRUM

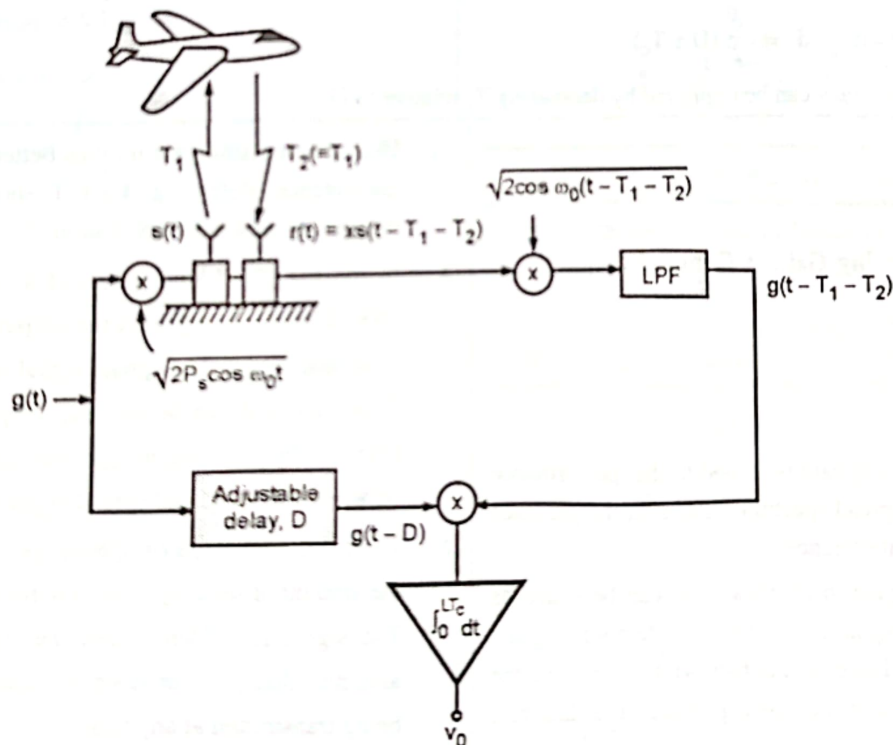


Fig. 4.7.1 : Ranging using DS spread spectrum

- Ranging is the application of DS-SS technology.
- Refer Fig. 4.7.1. It shows Ranging.
- In this DS signal $s(t) = \sqrt{2 P_s} g(t) \cos \omega_0 t$ is transmitted.
- The signal is reflected from the target. After $(T_1 + T_2)$ seconds it is received as $r(t)$ which is given as :

$$\begin{aligned} r(t) &= \alpha s(t - T_1 - T_2) \\ &= \alpha \sqrt{2 P_s} g(t - T_1 - T_2) \cos(\omega_0 t + \theta) \end{aligned}$$

where α = signal attenuation

θ = phase due to time delay = $-\omega_0 (T_1 + T_2)$

- Carrier $\{\sqrt{2} \cos \omega_0 (t - T_1 - T_2)\}$ is used and PN sequence $g(t - T_1 - T_2)$ is extracted.
- This extracted PN is correlated with $g(t - D)$. Here $g(t - D)$ is the PN waveform delayed by known time D . Hence, let $R_g(D)$ be the correlation function which is also output of integrator. It is given as,

$$v_0 = R_g(D) = \int_0^{LT_c} g(t - T_1 - T_2) g(t - D) dt$$

Where limits of integral shows length of PN sequence.

- $R_g(D)$ is negligibly small if the difference $[D - (T_1 + T_2)]$ exceeds the chip duration. $R_g(D)$ is maximum for $D = T_1 + T_2$. Thus the range is determined by adjusting D to maximize v_0 .
- For $T_1 = T_2$ and $c = 3 \times 10^8$ m/s, $d = \frac{1}{2} cD$
- The precision of the measurement is the chip duration $\pm T_c$. It corresponds to a distance $c T_c/2$ so that,

$$d = \frac{c}{2} (D \pm T_c)$$

and measured accuracy can be improved by decreasing T_c relative to D .

4.8 PROCESSING GAIN

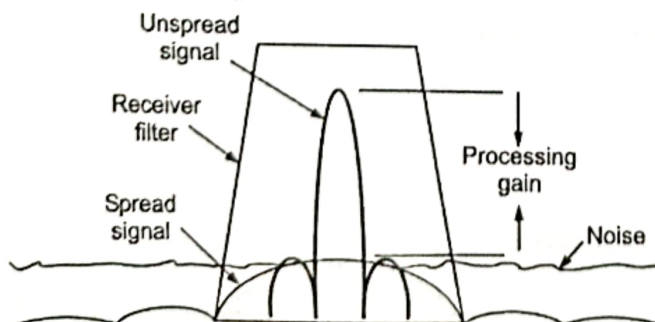
4.8.1 Processing Gain : Concept

UQ. Explain in Brief : Processing Gain.

SPPU - (E&TC-Sem. 5) May 15, 2 Marks

Ans. :

- The term processing gain is related to the performance measure of the spread spectrum system in the presence of narrowband interference.
- The processing gain in DSSS system can be stated as the ratio of the signal to jammer (interference) signals at the receiver. Hence, as the bandwidth increases, the processing gain also increases, provided the data rate remains constant.



(1F14)Fig. 4.8.1 : Concept of processing gain in DSSS system

- Higher processing gain implies better immunity against interference. Refer Fig. 4.8.1. It shows the concept of processing gain in DSSS system.
- In the Fig. 4.8.1, the unspread signal is narrowband PSK. The spread signal is the output of multiplier, i.e., wideband signal. The spread signal is close to the noise floor. Hence it can be said as, the processing gain for DSSS is the jamming margin that shows the difference between outspread and spread signals.
- The main advantage of spread spectrum technology is the amount of security against narrowband interference. The signal is hidden within the signal space as it is assumed that jammer does not know which signal is being transmitted at any time.
- The only thing done by jammer is to jam the entire space with its fixed total power and hence, each signal co-ordinate will have limited amount of interference or it also jams a portion of signal space with its total power, thus remaining signal space is free of interference.
- In all, it is necessary to develop signal space representations of the transmitted signal and interfering signal i.e. jamming signal.

4.8.2 Derivation of Processing Gain

GQ. Derive the processing gain of DS/BPSK system.

Ans. :

- Let the set of orthonormal basis functions :

$$\phi_k(t) = \begin{cases} \sqrt{\frac{2}{T_c}} \cdot \cos(2\pi f_c t) & \dots k T_c \leq t \leq (k+1) T_c \dots (4.8.1) \\ \dots \text{otherwise} \end{cases}$$

$$\bar{\phi}_k(t) = \begin{cases} \sqrt{\frac{2}{T_c}} \cdot \sin(2\pi f_c t) & \dots k T_c \leq t \leq (k+1) T_c \dots (4.8.2) \\ \dots \text{otherwise} \end{cases}$$

where, $[k = 0, 1, \dots, N-1]$

T_c = Chip duration

f_c = Carrier frequency

N = Number of chips per bit

From the Equation (4.5.1),

$$\begin{aligned} x(t) &= c(t) \cdot s(t) \\ &= \pm \sqrt{\frac{2E_b}{T_b}} \cdot \cos(2\pi f_c t) \cdot c(t) \dots (4.8.3) \end{aligned}$$

Since $s(t) = A_c \cos 2\pi f_c t$

$$= \sqrt{\frac{2E_b}{T_b}} \cdot \cos 2\pi f_c t$$

But $T_b = N T_c$, putting it in Equation (4.8.3)

$$\text{we get, } x(t) = \pm \sqrt{\frac{2E_b}{N T_c}} \cos(2\pi f_c t) \cdot c(t)$$

$$\therefore x(t) = \pm \sqrt{\frac{E_b}{N}} \sum_{k=0}^{N-1} C_k \phi_k(t), \quad 0 \leq t \leq T_b$$

...from Equation (4.8.1)...(4.8.4)

where, E_b = signal energy per bit

'+' sign, i.e., +1 = logic 1

'-' sign, i.e., -1 = logic 0

$$C_k = \{c_0, c_1, c_2 \dots c_{N-1}\}$$

= PN code sequence and C_k takes value ± 1 .

- Hence it can be said as, $x(t)$ is N-dimensional signal, and it needs 'N' orthonormal functions for representation. Now we will represent $j(t)$ which represents jammer signal.

- Generally, it is assumed that jammer distributes all its energy equally in N-dimensional signal space across $x(t)$. Hence, jammer is represented in general form as,

$$j(t) = \sum_{k=0}^{N-1} j_k \phi_k(t) + \sum_{k=0}^{N-1} \tilde{j}_k \bar{\phi}_k(t), \quad \dots 0 \leq t \leq T_b \dots (4.8.5)$$

where,

$$j_k = \int_0^{T_b} j(t) \phi_k(t) dt, \quad k = 0, 1, \dots, N-1 \dots (4.8.6)$$

similarly

$$\tilde{j}_k = \int_0^{T_b} j(t) \bar{\phi}_k(t) dt, \quad k = 0, 1, \dots, N-1 \dots (4.8.7)$$

- Therefore, jamming signal $j(t)$ is 2N-dimensional. Now, the average power of jamming interference signal $j(t)$ is given by,

$$\begin{aligned} J &= \frac{1}{T_b} \int_0^{T_b} j^2(t) dt \\ \therefore J &= \frac{1}{T_b} \sum_{k=0}^{N-1} j_k^2 + \frac{1}{T_b} \sum_{k=0}^{N-1} \tilde{j}_k^2 \dots (4.8.8) \end{aligned}$$

- But as per assumption, jamming signal distributes energy equally in signal space of $x(t)$, i.e., in the cosine and sine co-ordinates in Equation (4.8.6) and (4.8.7). Therefore assuming,

$$\sum_{k=0}^{N-1} \tilde{j}_k^2 = \sum_{k=0}^{N-1} j_k^2$$

- Putting this in Equation (4.8.8) we get,

$$J = \frac{2}{T_b} \sum_{k=0}^{N-1} j_k^2 \dots (4.8.9)$$

- For determining SNR at the input and output of the receiver, coherent detector's output should be considered. To express coherent detector's output, two components should be considered namely:

(i) v_s = output due to spread binary PSK, $s(t)$.

(ii) v_{cj} = output due to spread interference $c(t) j(t)$.

- These output components can be defined as,

$$(i) \quad v_s = \sqrt{\frac{2}{T_b}} \int_0^{T_b} s(t) \cos(2\pi f_c t) dt \dots (4.8.10)$$

$$(ii) \quad v_{cj} = \sqrt{\frac{2}{T_b}} \int_0^{T_b} c(t) j(t) \cos(2\pi f_c t) dt \dots (4.8.11)$$

$$\text{Therefore, } V = v_s + v_{cj} \dots (4.8.12)$$

As we know,

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$$s(t) = \pm \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t), 0 \leq t \leq T_b$$

where, +1 = logic 1

and -1 = logic 0

In this, f_c is integral multiple of $\frac{1}{T_b}$. Hence,

$$v_s = \pm \sqrt{E_b} \quad \text{where } E_b = \text{bit energy} \quad \dots(4.8.13)$$

Similarly,

$$v_{cj} = \sqrt{\frac{2}{T_b}} \sum_{k=0}^{N-1} C_k \int_{kT_c}^{(k+1)T_c} j(t) \cos(2\pi f_c t) dt$$

Adjusting the terms we get,

$$v_{cj} = \sqrt{\frac{2}{T_b}} \times \sqrt{\frac{T_c}{2}} \sum_{k=0}^{N-1} C_k \int_{kT_c}^{(k+1)T_c} j(t) \sqrt{\frac{2}{T_c}} \cos(2\pi f_c t) dt$$

$$\therefore v_{cj} = \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k \int_0^{T_b} j(t) \phi_k(t) dt$$

...[kT_c to $(k+1)T_c$ is one bit duration]

$$\therefore v_{cj} = \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k j_k \quad \dots(4.8.14)$$

- PN sequence is independent and identically distributed (i. i. d) binary sequence. Therefore, defining V_{cj} and C_k random variables with sample values v_{cj} and c_k , respectively.

$$\therefore V_{cj} = \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k j_k \quad \dots(4.8.15)$$

- Assuming probability of occurring logic '0' and logic '1' (+1) as $\frac{1}{2}$ each we get, $C_k = \pm 1$

$$\therefore P(C_k = 1) = P(C_k = -1) = \frac{1}{2} \quad \dots(4.8.16)$$

- Determining mean of random variable,

$$\begin{aligned} E[C_k j_k | j_k] &= j_k P(C_k = 1) - j_k P(C_k = -1) \\ &= \frac{1}{2} j_k - \frac{1}{2} j_k = 0 \quad \dots(4.8.17) \end{aligned}$$

Variance

$$\text{Var}[V_{cj} | j] = \frac{1}{N} \sum_{k=0}^{N-1} j_k^2 \quad \dots(4.8.18)$$

But $J = \frac{2}{T_b} \sum_{k=0}^{N-1} j_k^2$, and $N = \frac{T_b}{T_c}$

$$\therefore \text{Var}[V_{cj} | j] = J \times \frac{T_b}{2} \times \frac{T_c}{T_b}$$

$$\therefore \text{Var}[V_{cj} | j] = \frac{JT_c}{2} \quad \dots(4.8.19)$$

- SNR at the output is the ratio of signal power to noise variance. As already shown,

Signal energy = E_b .

$$\begin{aligned} \therefore \text{SNR}_o &= \frac{E_b}{\text{Var}[V_{cj} | j]} \\ &= \frac{E_b}{\frac{JT_c}{2}} = \frac{2E_b}{JT_c} \quad \dots(4.8.20) \end{aligned}$$

$$\text{Input SNR} = \frac{\text{average signal power at the receiver input}}{\text{average interference power}}$$

$$\text{SNR}_o = \frac{E_b/T_b}{J} \quad \dots(4.8.21)$$

Eliminating E_b/J from Equation (4.8.20) and (4.8.21), we get

$$(\text{SNR})_o = \frac{2T_b}{T_c} (\text{SNR})_i \quad \dots(4.8.22)$$

Processing gain is defined as the gain in SNR obtained by the use of spread spectrum.

$$\therefore \text{Processing gain, } P_G = \frac{T_b}{T_c} \quad \dots(4.8.23)$$

It refers to the gain achieved by processing a spread spectrum signal over an un-spread signal.

$$\therefore 10 \log_{10} (\text{SNR})_o = 10 \log_{10} (\text{SNR})_i + 3 + 10 \log_{10} (P_G) \text{ dB} \quad \dots(4.8.24)$$

- The term 3dB in Equation (4.8.24) is because of the gain in SNR obtained through the use of coherent detection. The longer the PN sequence, smaller is the chip time T_c and larger will be the processing gain. The processing gain can also be determined in terms of bit rate $\bar{R}_b$ and chip rate R_c .

$$R_b = \frac{1}{T_b}$$

$$\text{and } R_c = \frac{1}{T_c} \text{ hence,}$$

$$\text{Processing gain } P_G = \frac{R_c}{R_b}$$

4.9 PROBABILITY OF ERROR OF DS/BPSK

GQ. Derive the probability of error for DS/BPSK system.

$$\text{Hint: } P_e \approx \frac{1}{2} \operatorname{erfc} \left[\sqrt{\frac{E_b}{JT_c}} \right]$$

☒ **Ans. :**

- As we know, $V = v_s + v_{cj}$...from Equation (4.8.12)

$$\text{And } v_s = \pm \sqrt{E_b} \quad \dots \text{from Equation (4.8.13)}$$

$$\therefore V = \pm \sqrt{E_b} + v_{cj} = \pm \sqrt{E_b} + V_{cj}$$

where, E_b = Signal energy per bit

and V_{cj} = Sample value of equivalent noise component v_{cj}

- '+' sign refers to +1, i.e., logic '1' and '-' sign refers to -1, i.e., logic '0'. At the receiver, decision device checks for the threshold value. If the detector's output is greater than threshold of zero, then the bit is detected as '1' and if it is less, then zero. If the detector output is equal to exact zero, the decision device makes a random decision.

Both bits are assumed to have equal probability. Hence, average probability of error is same as that of conditional probability.

$$\therefore P_e = P(V > 0 | \text{symbol '0' was sent})$$

$$\therefore P_e = P(V_{cj} > \sqrt{E_b}) \quad \dots (4.9.1)$$

- Hence, probability of error P_e depends on the random variable V_{cj} where,

$$V_{cj} = \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k j_k$$

- Central limit theorem states that under very general conditions, the probability distribution of the sum of j statistically independent random variables approaches the Gaussian distribution as $j \rightarrow \infty$, independent of individual distribution functions.

Hence, from central limit theorem, it can be stated that for large N , V_{cj} tends to have Gaussian distribution. In PN sequence, N is usually large in DSSS.

- Hence, V_{cj} can be approximated as Gaussian random variable with zero mean and its variance will be $\frac{JT_c}{2}$

Therefore probability of error for $V_{cj} > \sqrt{E_b}$ and average probability thereby can be expressed as,

$$P_e \approx \frac{1}{2} \operatorname{erfc} \left[\sqrt{\frac{E_b}{JT_c}} \right]$$

This is average probability for DS/BPSK with large spread factor N .

4.9.1 Antijam Characteristics/Jamming Margin

UQ. Explain in brief: Jamming Margin.

SPPU - (E&TC-Sem. 5) May 15, 2 Marks

☒ **Ans. :**

The error probability of coherent BPSK signal is given as,

$$P_e = \frac{1}{2} \operatorname{erfc} \left[\sqrt{\frac{E_b}{N_0}} \right] \quad \dots (4.9.2)$$

And error probability of DS/BPSK signal is given as,

$$P_e = \frac{1}{2} \operatorname{erfc} \left[\sqrt{\frac{E_b}{JT_c}} \right] \quad \dots (4.9.3)$$

Comparing Equations (4.9.2) and (4.9.3) we get,

$$N_0 = JT_c \quad \dots (4.9.4)$$

$$\text{Now energy per bit, } E_b = PT_b \quad \dots (4.9.5)$$

where P = average signal power

T_b = bit duration

$$\therefore \frac{E_b}{N_0} = \frac{PT_b}{JT_c} \quad [\text{from Equations (4.9.4) and (4.9.5)}]$$

$$= \left(\frac{T_b}{T_c} \right) \left(\frac{P}{J} \right)$$

$$\text{But } P_G = \frac{T_b}{T_c}$$

$$\text{Hence, } \frac{E_b}{N_0} = P_G \times \left[\frac{P}{J} \right]$$

$$\therefore \frac{J}{P} = \frac{P_G}{E_b/N_0}$$

The ratio $\frac{J}{P}$ is nothing but the jamming margin.

Expressing jamming margin in dB,

$$[\text{Jamming margin}]_{\text{dB}} = [\text{Processing gain}]_{\text{dB}} - 10 \log_{10} \left[\frac{E_b}{N_0} \right]_{\text{min}}$$

Important Formulae

DSSS

1. Processing gain, $P_G = \frac{T_b}{T_c} = \frac{\text{Bit duration}}{\text{Chip duration}}$
2. $10 \log_{10} (\text{SNR})_o = 10 \log_{10} (\text{SNR})_i + 3 + 10 \log_{10} (P_G)$
3. Processing gain = $P_G = G_p = \frac{R_c}{R_b}$

Where $R_c = \frac{1}{T_c} = \frac{1}{\text{Chip duration}} = \text{Chip rate}$

$$R_b = \frac{1}{T_b} = \frac{1}{\text{Bit duration}} = \text{bit rate}$$

4. Probability of error for DS/BPSK signal is

$$P_e \approx \frac{1}{2} \text{erfc} \left[\sqrt{\frac{E_b}{JT_c}} \right]$$

Where, E_b = Bit energy

J = Average power of jamming interference signal

T_c = Chip duration

5. Jamming margin = $\frac{J}{P} = \frac{P_G}{E_b/N_0}$

Where P = Average signal power

P_G = Processing gain

$N_0 = JT_c$

6. Jamming margin in dB

$$[\text{Jamming margin}]_{\text{dB}} = [\text{Processing gain}]_{\text{dB}} - 10 \log_{10} \left[\frac{E_b}{N_0} \right]_{\text{min}}$$

7. $T_b = NT_c$

Where N = Number of chips per bit.

4.9.2 Examples on DSSS

Ex. 4.9.1 : A DSSS system has a 1.2288 mega chips per second (Mcps) code clock rate and a 9.6 kbps information rate. Calculate the processing gain. How much improvement in information rate is achieved if the code generation rate is changed to 5 Mcps and the processing gain to 256 ?

✓ Soln. :

Given : 1. $R_b = 9.6 \text{ kbps}$

2. $R_c = 1.2288 \times 10^6 \text{ chips per second}$

I) Processing gain $PG = G_p = \frac{R_c}{R_b}$

$$\therefore G_p = \frac{1.2288 \times 10^6}{9.6 \times 10^3} = 128$$

$$[G_p]_{\text{dB}} = 10 \log_{10} (128) = 21 \text{ dB}$$

II) New $G_p = 256$ and $R_c = 5 \text{ Mcps}$

$$\therefore G_p = \frac{R_c}{R_b}$$

$$\therefore R_b = \frac{R_c}{G_p} = \frac{5 \times 10^6}{256} = 19.53 \text{ kbps}$$

$\therefore$ Improvement in information rate

$$= 19.53 \text{ kbps} - 9.6 \text{ kbps}$$

$$= 9.93 \text{ kbps}$$

UEx. 4.9.2 SPPU - (E & TC-Sem. 5) Dec. 14, Dec. 16,

Dec. 17, 6 Marks

A spread spectrum communication system has the following parameters. Information bit duration, $T_b = 4.095 \text{ ms}$, PN chip duration, $T_c = 1 \mu\text{s}$. Find the processing gain. What is the number of shift registers required? Also find the jamming margin if $E_b/N_0 = 10$ for BPSK scheme.

✓ Soln. :

Given : 1. $T_b = 4.095 \text{ ms}$ 2. $T_c = 1 \mu\text{s}$ 3. $\frac{E_b}{N_0} = 10$

► Step 1 :

$$\text{Processing gain } PG = G_p = \frac{T_b}{T_c} = \frac{4.095 \times 10^{-3}}{1 \times 10^{-6}} = 4095$$

Hence $G_p = 4095$

► Step 2 :

$$T_b = NT_c$$

$$\therefore 4.095 \times 10^{-3} = N \times 1 \times 10^{-6}$$

$$\therefore N = 4095$$

$\therefore$ The number of shift registers required is given as,

$$N = 2^m - 1 \quad \therefore 4095 = 2^m - 1$$

$$\therefore 2^m = 4096 \dots \text{Taking log on both sides,}$$

$$\therefore m \log 2 = \log 4096 \quad \therefore m = 12$$

► Step 3 :

$$\begin{aligned}
 [\text{Jamming margin}]_{\text{dB}} &= [G_P]_{\text{dB}} - 10 \log_{10} \left[\frac{E_b}{N_0} \right] \\
 &= 10 \log 4095 - 10 \log (10) \\
 &= 36.1 - 10 = 26.1 \text{ dB}
 \end{aligned}$$

UEx. 4.9.3 SPPU - (E & TC-Sem. 5) May 16, 9 Marks

The bit duration in DS SS BPSK communication system is 4 ms and the chipping rate is 1 Mbps. Considering average error probability of 10^{-5} for detecting the message signal, calculate the processing gain and Jamming margin. Given $Q(4.25) = 10^{-5}$.

✓ Soln. :

Given :

$$\begin{aligned}
 (1) \quad T_b &= 4\text{ms} = 4 \times 10^{-3} \text{ s} & (2) \quad T_c &= 1\text{Mbps} = 10 \times 10^6 \\
 (3) \quad P_e &= 10^{-5} & (4) \quad Q(4.25) &= 10^{-5}
 \end{aligned}$$

► Step 1 :

$$\text{Processing gain } P_G = G_P = \frac{T_b}{T_c} = \frac{4 \times 10^{-3}}{1 \times 10^{-6}} = 4000$$

► Step 2 :

$$\begin{aligned}
 P_e &= \frac{1}{2} \text{erfc} \sqrt{\frac{E_b}{N_0}} = \frac{1}{2} \times 2Q \left[\sqrt{\frac{2E_b}{N_0}} \right] \\
 &\quad \dots (\text{Error probability of BPSK system}) \\
 \therefore 10^{-5} &= Q \left[\sqrt{\frac{2E_b}{N_0}} \right]
 \end{aligned}$$

$$\text{But } Q(4.25) = 10^{-5} \dots (\text{given}) \quad \therefore \sqrt{\frac{2E_b}{N_0}} = 4.25$$

Squaring on both sides we get,

$$\therefore \frac{2E_b}{N_0} = (4.25)^2 = 18.0625$$

$$\therefore \frac{E_b}{N_0} = 9.0312$$

► Step 3 :

$$[\text{Jamming margin}]_{\text{dB}} = [G_P]_{\text{dB}} - 10 \log_{10} \left[\frac{E_b}{N_0} \right]$$

$$\begin{aligned}
 \therefore [\text{Jamming margin}]_{\text{dB}} &= 10 \log (4000) - 10 \log (9.0312) \\
 &= 36.020 - 9.5574 \\
 &= 26.4626 \text{ dB}
 \end{aligned}$$

UEx. 4.9.4 SPPU - (E & TC-Sem. 5) May 18, 6 Marks

The information bit duration in DS-BPSK spread spectrum communication system is 5 ms while the chipping rate is 1 MHz. Assuming an average error probability of 10^{-5} for proper detection of message signal, calculate Jamming margin. Given $Q(4.25) = 10^{-5}$.

✓ Soln. :

$$\begin{aligned}
 \text{Given : (i) } T_b &= 5\text{ms} = 5 \times 10^{-3} \quad (\text{ii) } T_c = 1 \text{ MHz} = 1 \times 10^6 \\
 (\text{iii) } P_e &= 10^{-5} \quad (\text{iv) } Q(4.25) = 10^{-5}
 \end{aligned}$$

► Step 1 :

$$P_G = \frac{T_b}{T_c} = \frac{5 \times 10^{-3}}{1 \times 10^{-6}} = 5000$$

► Step 2 :

$$\begin{aligned}
 P_e &= \frac{1}{2} \text{erfc} \sqrt{\frac{E_b}{N_0}} = \frac{1}{2} \times 2Q \left[\sqrt{\frac{2E_b}{N_0}} \right] \\
 &\quad \dots (\text{Error probability of BPSK system})
 \end{aligned}$$

$$10^{-5} = Q \left[\sqrt{\frac{2E_b}{N_0}} \right]$$

$$\text{But, } Q(4.25) = 10^{-5} \quad \therefore \sqrt{\frac{2E_b}{N_0}} = 4.25$$

$$\therefore \frac{E_b}{N_0} = 9.0312$$

► Step 3 :

$$\begin{aligned}
 [\text{Jamming margin}]_{\text{dB}} &= [G_P]_{\text{dB}} - 10 \log_{10} \left[\frac{E_b}{N_0} \right]_{\text{min}} \\
 &= 10 \log (5000) - 10 \log (9.0312) \\
 &= 36.9897 - 9.5574 \\
 &= 27.4323 \text{ dB}
 \end{aligned}$$

Ex. 4.9.5 : The signal power received for desired signal of 100 kb/s bit rate signal is 1 mW. The chip frequency used is 100 MHz. A jamming signal is employed at the carrier frequency, the received power of which is 1W. Find (a) processing gain (b) error probability without jamming and with jamming. Noise power spectral density = 10^{-9} W/Hz.

✓ Soln. :

$$\begin{aligned}
 \text{Given : (i) } f_b &= 10^5 \text{ Hz} \quad (\text{ii) } f_c = 10^8 \\
 (\text{iii) } P_s &= 0.001 \quad (\text{iv) } J = 1 \\
 (\text{v) } N_0 &= 2 \times 10^{-9} \text{ W/Hz}
 \end{aligned}$$

(a) Processing gain

$$P_G = \frac{R_c}{R_b} = \frac{f_c}{f_b} = \frac{10^8}{10^5} = 1000$$

$$\text{Bit interval } T = \frac{1}{f_b} = 10^{-5} \text{ sec}$$

$$\text{Energy per bit, } E_b = P_s T_b = 0.001 \times 10^{-5} = 10^{-8}$$

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(b) Error probability

(i) Without jamming

$$P_e = \frac{1}{2} \operatorname{erfc} \sqrt{\frac{E_b}{N_0}} = 0.00078$$

(ii) With jamming

$$\begin{aligned} P_e &= \frac{1}{2} \operatorname{erfc} \sqrt{\frac{2 P_s \cdot P_G}{J}} \\ &= \frac{1}{2} \operatorname{erfc} \sqrt{\frac{2 \times 10^{-3} \times 1000}{1}} \\ &= 0.02307 \end{aligned}$$

Ex. 4.9.6 : A DS spread spectrum system of chip rate 10 MHz is used for ranging. If the reflected wave is received after 0.1 msec. Find the probable distance of the target.

✓ **Soln. :**

Given : (i) $f_c = 10 \text{ MHz}$ (ii) $D = 0.1 \text{ m sec.}$

$$\text{Chip period } T_c = \frac{1}{f_c} = \frac{1}{10 \times 10^6} = 10^{-7} \text{ seconds}$$

$$\text{Now estimated distance } d = \frac{1}{2} cD$$

$$\begin{aligned} d &= \frac{1}{2} \times 3 \times 10^8 \times 0.1 \times 10^{-3} \\ &= 15 \text{ km} \end{aligned}$$

$$\begin{aligned} \text{The tolerance } d &= \frac{CT_c}{2} \\ &= \frac{1}{2} \times 3 \times 10^8 \times 10^{-7} = 15 \text{ m} \end{aligned}$$

Thus the target is between $15000 - 15 = 14985 \text{ m}$ and $15000 + 15 = 15015 \text{ m}$ of the source.

4.10 FREQUENCY HOPPING SPREAD SPECTRUM (FHSS)

UQ. Draw and explain FHSS spread spectrum system with transmitter and receiver section.

SPPU - (E&TC-Sem. 5) Dec. 14, May 15, 8 Marks

UQ. Draw the block diagram of FH-SS system's transmitter and receiver. Write the functional names inside the blocks and input output signals of each block.

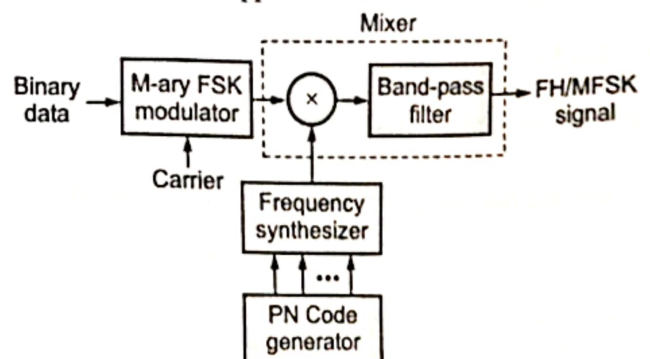
SPPU - (E&TC-Sem. 5) Dec. 16, 6 Marks

UQ. Draw the block diagram of FHSS transmitter and receiver and explain the various blocks.

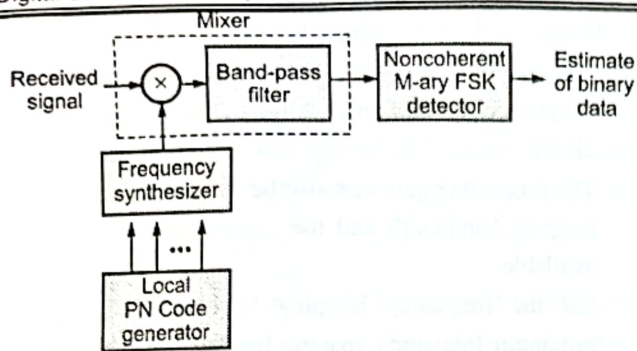
SPPU - (E&TC-Sem. 5) Dec. 17, 8 Marks

✓ **Ans. :**

1. The transmission method in which the carrier hops rapidly between the available channels according to the hopping pattern generated by PN sequence is called as 'frequency hopped spread spectrum'.
2. The allocated spectrum is divided into number of narrow bandwidth channels.
3. Generally, the modulation format used is **M-ary FSK (MFSK)**. The MFSK signal switches between the carrier's frequencies available pseudo randomly. These carriers are generated using the frequency synthesizer using the hopping sequence provided to it. The combination of the MFSK and FHSS techniques is simply called as FH/MFSK.
4. **Working principle of FHSS**
Refer Fig. 4.10.1(a) and (b). This system is actually two step modulation processes. Data modulation and frequency hopping modulation is carried out in it. For data modulation purpose the MFSK transmitter is used, and frequency hopping is carried out by the frequency synthesizer controlled by PN code.
5. In this the data symbol modulates the carrier whose frequency is determined by the hopping sequence generated by PN code.
6. Initially the binary user data is given as input to the MFSK transmitter. The modulated output of the MFSK transmitter then applied to the mixer.



(IF15) Fig. 4.10.1(a) : Block diagram of FHSS transmitter



(1F16)Fig. 4.10.1(b) : Block diagram of FHSS receiver

7. To the mixer, the other input is provided from the digital frequency synthesizer. The digital frequency synthesizer generates the carriers as per the hopping pattern generated by the PN code generator.
8. The output of the mixer is then passed to the bandpass filter. The filter is so designed that it selects the sum frequency components resulting from the multiplication process. The successive k bits of PN code drives the frequency synthesizer. Hence, frequency synthesizer is able to generate 2^k different carrier frequencies.
9. As large bandwidth of few GHz range is required for FHSS technique, coherent detection is possible only within each hop. This is because; the frequency synthesizer is not able to maintain phase coherence over successive hops. Hence, non-coherent detection is used in this technology.
10. At the receiver, the signal is first mixed with the output of the frequency synthesizer. The output of the frequency synthesizer is controlled by the PN code generator. The PN code used at transmitter and receiver has to be same. Hence with the same key, PN code is generally locally generated at the receiver. Mixer removes the frequency hopping in the received signal.
11. The output of the mixer is given to the bandpass filter and then sent for the non-coherent detection.
12. The single tone carrier frequency is called as 'chip' in context with FHSS. Hence the chip rate is given by,

$$R_c = \max(R_h, R_s)$$

Where, R_c = chip rate; R_h = hopping rate

R_s = symbol rate

13. At each hop, the MFSK carrier tone frequencies are separated in frequency by an integral multiple of the chip rate, R_c . This indicates that the transmitted symbol will not produce adjacent channel interference (from one filter output to adjacent one) or spill over or crosstalk in other $M-1$ non-coherent matched filters in the MFSK receiver.

4.10.1 Advantages, Disadvantages and Applications of FHSS

UQ. What are advantages of FHSS?

SPPU - (E&TC-Sem. 5) Dec. 17, 2 Marks

UQ. Write advantages and disadvantages of FHSS.

SPPU - (E&TC-Sem. 5) May 18, 4 Marks

✓ Ans. :

Advantages of FHSS

1. It needs shorter acquisition time.
2. Robust technology.

Disadvantages of FHSS

1. It requires frequency synthesizers which increase complexity of the system.
2. Sharp cut-off filtering is needed at the receivers.

Applications of FHSS

1. WLAN (wireless LAN) standards
2. WPAN (wireless PAN) standards like Bluetooth, etc.

4.10.2 Processing Gain for FHSS

GQ. Derive processing gain for FHSS.

$$\text{Hint : } PG = G_p = \frac{B_w}{R_c} = 2^k$$

✓ Ans. :

- Let the jammer spreads its average power J over the entire frequency hopped spectrum. As explained in case of DSSS processing gain, the average power will be equivalent to an AWGN with power spectral density $N_0/2$.

Where, $N_0 = J/B_w$,

B_w = bandwidth of the FHSS system

Therefore, $\frac{E}{N_0} = \frac{P/J}{B_w/R_s}$

Where, P/J = reciprocal of the jamming margin

P = average signal power and

J = average interference power

B_w = FH bandwidth

R_s = symbol rate

- The processing gain for the FH/MFSK system is given by the ratio,

$$PG = G_p = \frac{B_w}{R_s} = 2^k$$

This means that the processing gain in dB is given by,

$$[10 \log_{10} 2^k]_{dB} = 10 \times k \log_{10} 2 = 3k \text{ dB}$$

Where, k = length of the PN sequence used

- If the jammer does not spread the power equally over entire spectrum, the processing gain would be less than $3k$ dB.
- The processing gain can also be determined in terms of hopping bandwidth and the carrier frequency spacing available.
- Let the frequency hopping bandwidth be B_w and minimum frequency spacing between consecutive hop position be Δf , then processing gain is given as,

$$G_p = \frac{\text{Hopping Bandwidth}}{\text{Minimum frequency spacing}}$$

$$G_p = \frac{M \cdot \Delta f}{\Delta f} = M \quad \dots (4.10.1)$$

4.10.3 Types of FHSS Technique

UQ. Explain in brief: (i) Fast Frequency Hopping (ii) Slow Frequency Hopping.

SPPU - (E&TC-Sem. 5) May 15, 4 Marks

UQ. Explain the i) Slow frequency hopping. ii) Fast frequency hopping spread spectrum with the help of relevant diagram.

SPPU - (E&TC-Sem. 5) May 17, 6 Marks

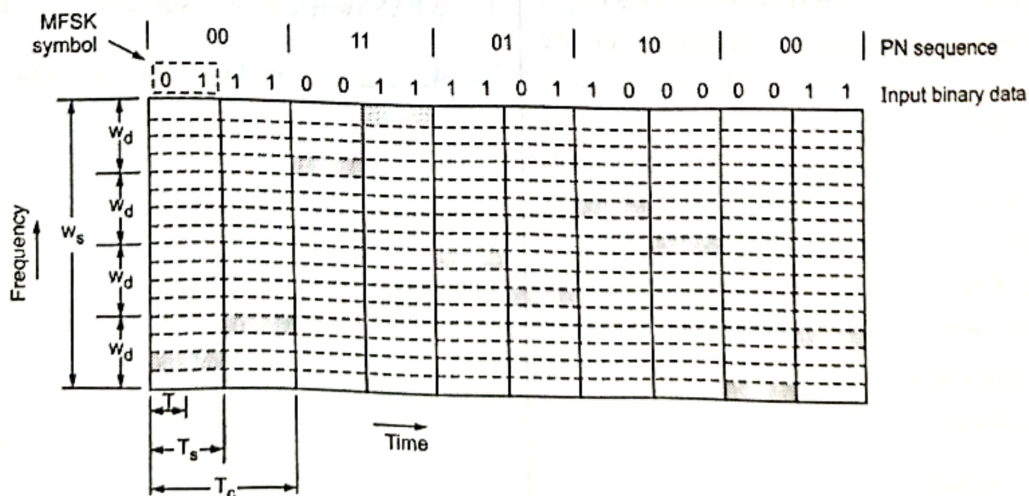
Ans. :

1. Slow FHSS
2. Fast FHSS

1. Slow FHSS (SFH)

Several modulation symbols are carried by single hop in slow FHSS. It means that it has 'N' (number of bits) symbols per hop. Hence, each symbol in slow FHSS is known as a chip.

In slow FHSS, $R_c = R_s = \frac{R_h}{K} \geq R_h$ where, $k = \log_2 M$



(1F17)Fig. 4.10.2(a) : Slow FHSS waveforms

- At each hop, the MFSK tone frequencies are separated by integer multiple of chip rate or symbol rate, when $R_c = R_s$. This separation is necessary to provide desired degree of orthogonality between the frequencies and it also allows

reliable non-coherent detection. Therefore, minimum bandwidth of an MFSK signal should be about $M R_s$. In this technique, the shortest uninterrupted waveform is nothing but the data symbol.

- **Dwell time:** The time for which the receiver stays on the same frequency. Slow FHSS system has longer Dwell time. In the Fig. 4.10.2(a), one symbol of 2 bits are transmitted using single carrier frequency. After every symbol duration, the receiver hops to another carrier to send next MFSK symbol.

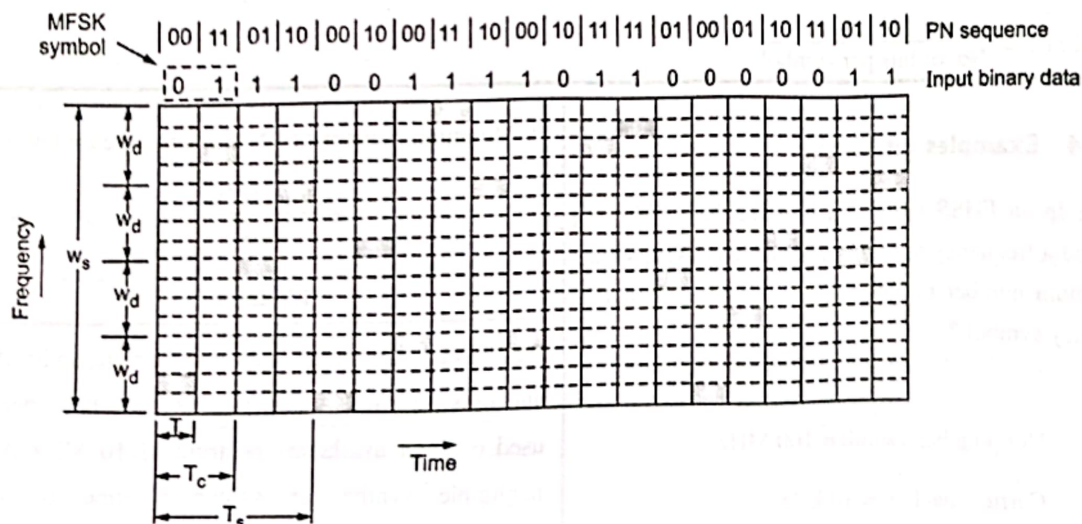
► 2. Fast FHSS (FFH)

- Several frequency hops are there to carry single symbol. It means that it has ' N ' hops per symbol. The shortest uninterrupted wave is that of a hop. Here, each hop is a **chip**.
- In fast FHSS system, MFSK is used in which $N = \log_2 M$ bits are used to determine which one of M frequencies should be used. [N = number of bits per symbol, M = number of different frequency tones used in MFSK]
- Generally jammers involve in two operations :
 - a) Measurement of the spectral components of the transmitted signals and
 - b) Retuning of the interfering signal to that portion of the frequency band.

Therefore, it is necessary that the transmitted signal must be hopped to new carrier frequency before the jammer signal completes its above said operations.

Refer Fig. 4.10.2 (b).

- Detection process is bit different in fast FHSS as compared to slow FHSS. For detection purpose, any one of the following **two methods** can be used :
 - a) To detect the de-hopped MFSK symbol, each symbol separate decision is made, and rule of majority vote is applied.
 - b) For each FH/MFSK symbol, likelihood functions are determined as a function of the total signal received over k chips, and the largest one is selected.



(1F18)Fig. 4.10.2(b) : Fast FHSS waveforms

- The detection method using the likelihood functions is more optimum as it reduces the probability of symbol error (E_b/N_0). In the Fig. 4.10.2(b), two carriers are used for transmission for one MFSK symbol. Hence, the hopping rate is more and the dwell time is less.

- Processing gain for fast FHSS

In Fast FHSS, Hopping Bandwidth = $(KM\Delta f)L$

Where K = frequency multiplication factor M = Number of carriers in MFSK system

N = Number of bits per symbol L = frequency hops per symbol

Hence process in gain,

$$G_p = \frac{(KM\Delta f)L}{\Delta f} \quad \therefore G_p = MKL$$

- Hence it can be said that the processing gain of a fast frequency hop system is dependent upon the number of frequencies used (M), the number of hops per symbol (L) and the frequency multiplication factor (K).

Important Formulae

1. Processing gain for FHSS $P_G = G_p = \frac{B_w}{R_s} = 2^K$

Where B_w = FH Bandwidth, R_s = Symbol rate, K = Length of PN sequence used

2. Processing gain in dB; $[G_p]_{dB} = 3K \text{ dB}$

3. $G_p = \frac{\text{Hopping Bandwidth}}{\text{Minimum Frequency Spacing}}$

4. Hopping bandwidth = $KM\Delta fL$

Where M = Level of FSK used ; Δf = Carrier frequency spacing ; L = Frequency hops per symbol

5. Processing gain $G_p = MKL$

6. Symbol rate = $\frac{\text{Bit rate}}{\text{No. of bits per symbol}}$

4.10.4 Examples on FHSS

Ex. 4.10.1 : In an FHSS system, a hopping bandwidth of 100 MHz and a frequency spacing of 10 KHz is used. What is the minimum number of PN chips that are required for each frequency symbol ?

☒ **Soln. :**

Given : 1. Hopping bandwidth = 100 MHz

2. Carrier spacing = 10 kHz

$$\begin{aligned} \text{Number of frequency tones in hopping bandwidth} &= \frac{\text{Hopping bandwidth}}{\text{carrier spacing}} \\ &= \frac{100 \times 10^6}{10 \times 10^3} = 10^4 \end{aligned}$$

$\therefore$ Minimum number of PN chips required = $\log_2 (10^4)$

$$= \frac{\log (10^4)}{\log 2}$$

$$= 13 \text{ chips}$$

Ex. 4.10.2 : A communication system transmits at 120 kbps and uses 32-FSK. A hop rate of 2000 hops per second is used over an available spectrum of 10 MHz Assuming a negligible synthesizer switching time between hops, calculate :

- Data symbol transmitted per hop.
- The number of non-overlapping hop frequencies.

✓ Soln. :

- Given : (i) $M = 32$
 (ii) Bit rate = 120 kbps
 (iii) hop rate = 2000 hops/sec
 (iv) spectrum available = 10 MHz

► Step 1 :

$$M = 2^N \quad \therefore 32 = 2^N$$

$$\therefore N = 5 \text{ bits per symbol}$$

► Step 2 :

$$\begin{aligned} \text{Symbol rate} &= \frac{\text{Bit rate}}{\text{No. of bits per symbol}} \\ &= \frac{120 \times 10^3}{5} \\ &= 24 \text{ kbps} \end{aligned}$$

Note : As symbol rate > hop rate, it is slow FHSS.

► Step 3 :

$$\begin{aligned} \text{Number of symbols per hop} &= \frac{\text{Symbol rate}}{\text{hop rate}} = \frac{R_s}{R_h} \\ &= \frac{24 \times 10^3}{2 \times 10^3} = 12 \end{aligned}$$

$$\therefore \text{Data symbols transmitted per hop} = 12$$

► Step 4 : Minimum bandwidth for MFSK.

$$\begin{aligned} B &= \text{No. of symbols per hop} \times \text{symbol rate} \\ &= 12 \times 24 \times 10^3 = 288 \text{ kHz} \end{aligned}$$

► Step 5 : Number of non-overlapping hop frequencies.

$$\begin{aligned} &= \frac{\text{available spectrum}}{\text{Bandwidth of MFSK system}} \\ &= \frac{10 \times 10^6}{288 \times 10^3} = 35 \end{aligned}$$

Ex. 4.10.3 : In an FHSS system in which the input data rate is 200 bits per second. The modulation scheme of 32-ary FSK is used to generate the modulation symbol. The frequency hopping rate is 200 hops per second. Calculate
 (a) minimum separation between frequency tones;
 (b) number of frequency tones produced by a frequency synthesizer (c) processing gain and (d) hopping bandwidth. Assume frequency multiplication factor $k = 1$.

✓ Soln. :

- Given : (i) $M = 32$
 (ii) Data rate = 200 bps
 (iii) Hopping rate = 200 hops/sec
 $M = 2^N \quad \therefore 32 = 2^N$
 $\therefore N = \text{Number of chips per symbols} = 5$

► Step 1 :

$$\begin{aligned} \text{Symbol rate } R_s &= \frac{\text{data rate}}{N} = \frac{200}{5} \\ &= 40 \text{ symbols/sec.} \end{aligned}$$

Here, hopping rate > symbol rate. Hence, the system is fast FHSS.

$$\text{► Step 2 : Symbol duration} = \frac{1}{40} = 25 \text{ ms}$$

► Step 3 :

$$\text{Frequency hops per symbol} = \frac{200}{40} = 5 \text{ hops/symbol}$$

► Step 4 :

$$\text{Chip duration} = \frac{25}{5} = 5 \text{ ms}$$

► Step 5 :

$$\begin{aligned} \text{Minimum separation between tones} &= \frac{1}{5 \times 10^{-3}} \\ &= 200 \text{ Hz} \end{aligned}$$

► Step 6 :

$$\begin{aligned} \text{Hopping bandwidth} &= \text{Frequency multiplication factor} \\ &\quad (K) \times M \times (\text{carrier Frequency separation} \times \text{frequency hops per symbol } (L)) \end{aligned}$$

$$\begin{aligned} \therefore \text{Hopping bandwidth} &= (KM\Delta f) L = 1 \times 32 \times 200 \times 5 \\ &= 64 \text{ kHz} \end{aligned}$$

$$\text{Processing gain } G_p = MKL = 32 \times 1 \times 5 = 160$$

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End Sem

UEx. 4.10.4 SPPU - (E & TC-Sem. 5) May 15, 8 Marks

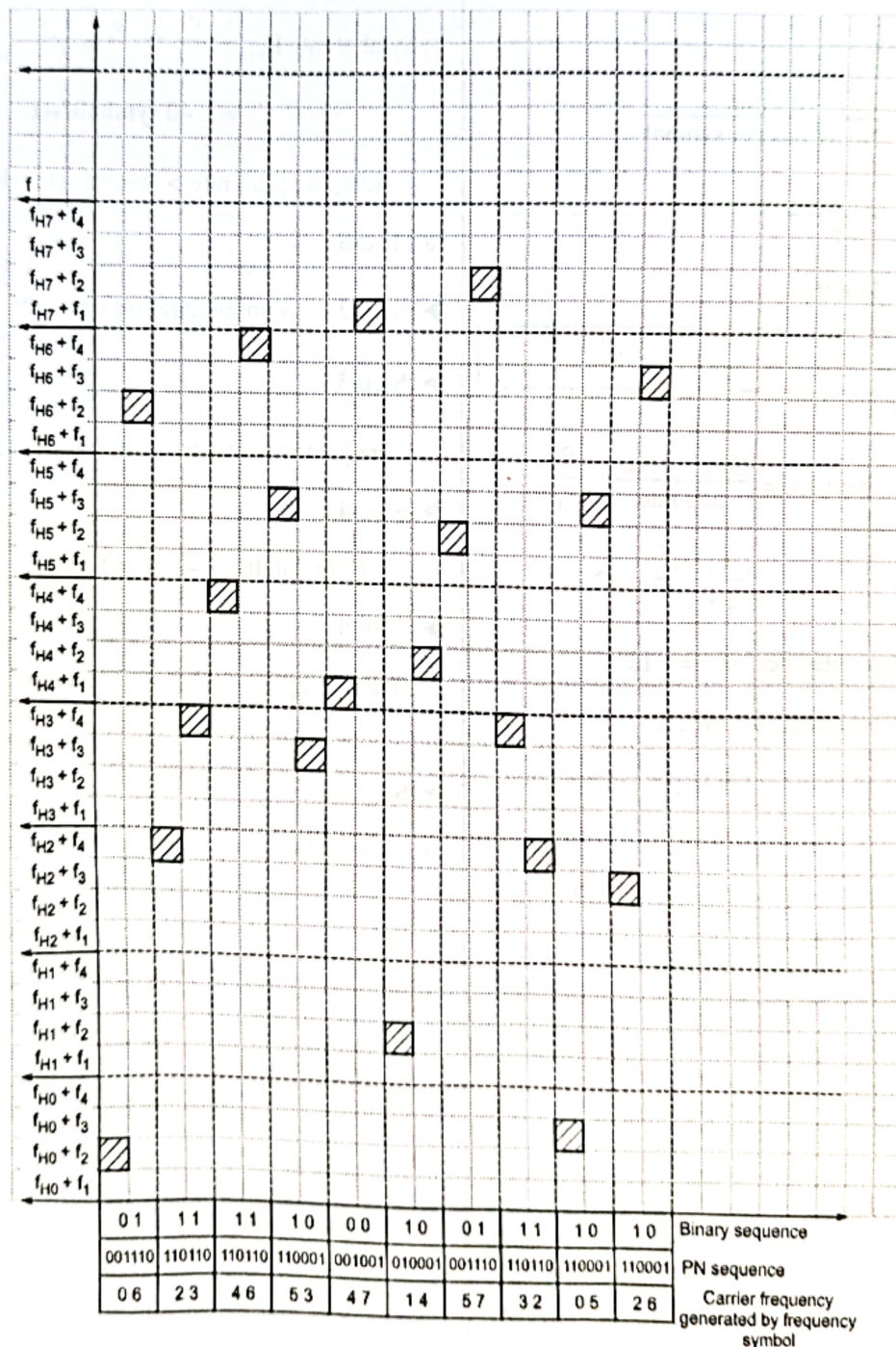
Represent variation of the frequency of a fast hop spread spectrum system with binary FSK, having following parameters:
 Number of bits per MFSK symbol $K = 2$; Number of MFSK tones $M = 2^K = 4$ Length of PN segment per hop $k = 3$;
 Total number of frequency hops $2^k = 8$. For the binary message of 01111110001001111010

✓ Soln. :

00	→	f_1
01	→	f_2
10	→	f_3
11	→	f_4

MFSK symbol

Carrier allocation for bits in MFSK symbol



(1F21) Fig. P. 4.10.4

UEx. 4.10.5 SPPU - (E & TC-Sem. 5) Dec. 15, 8 Marks

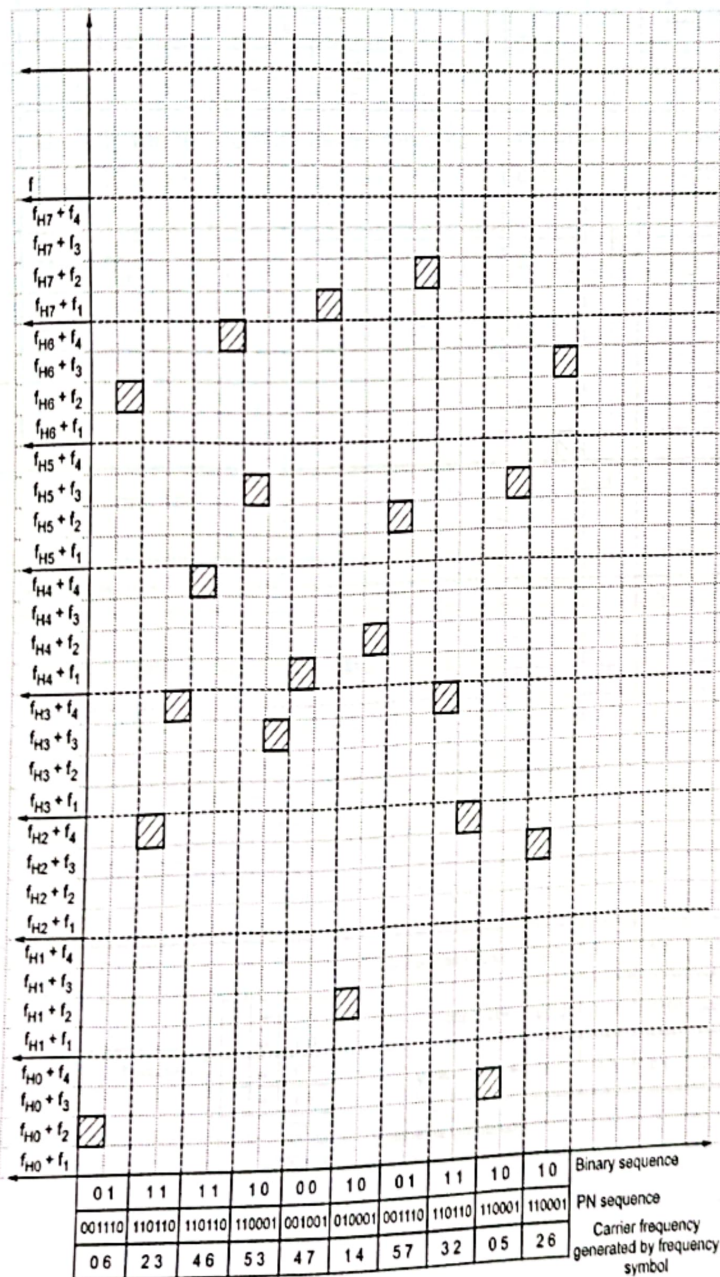
The signal has the following parameter number of bits per MFSK symbol $K = 2$, number of MFSK tone $M = 2^K = 4$ length of PN sequence per hop $K = 3$, total number frequency hops $2^K = 8$. Sketch the output transmits frequency of fast FH/MFSK signals.

✓ Soln. :

00	→	f_1
01	→	f_2
10	→	f_3
11	→	f_4

Carrier allocation for bits in MFSK symbol

MFSK symbol



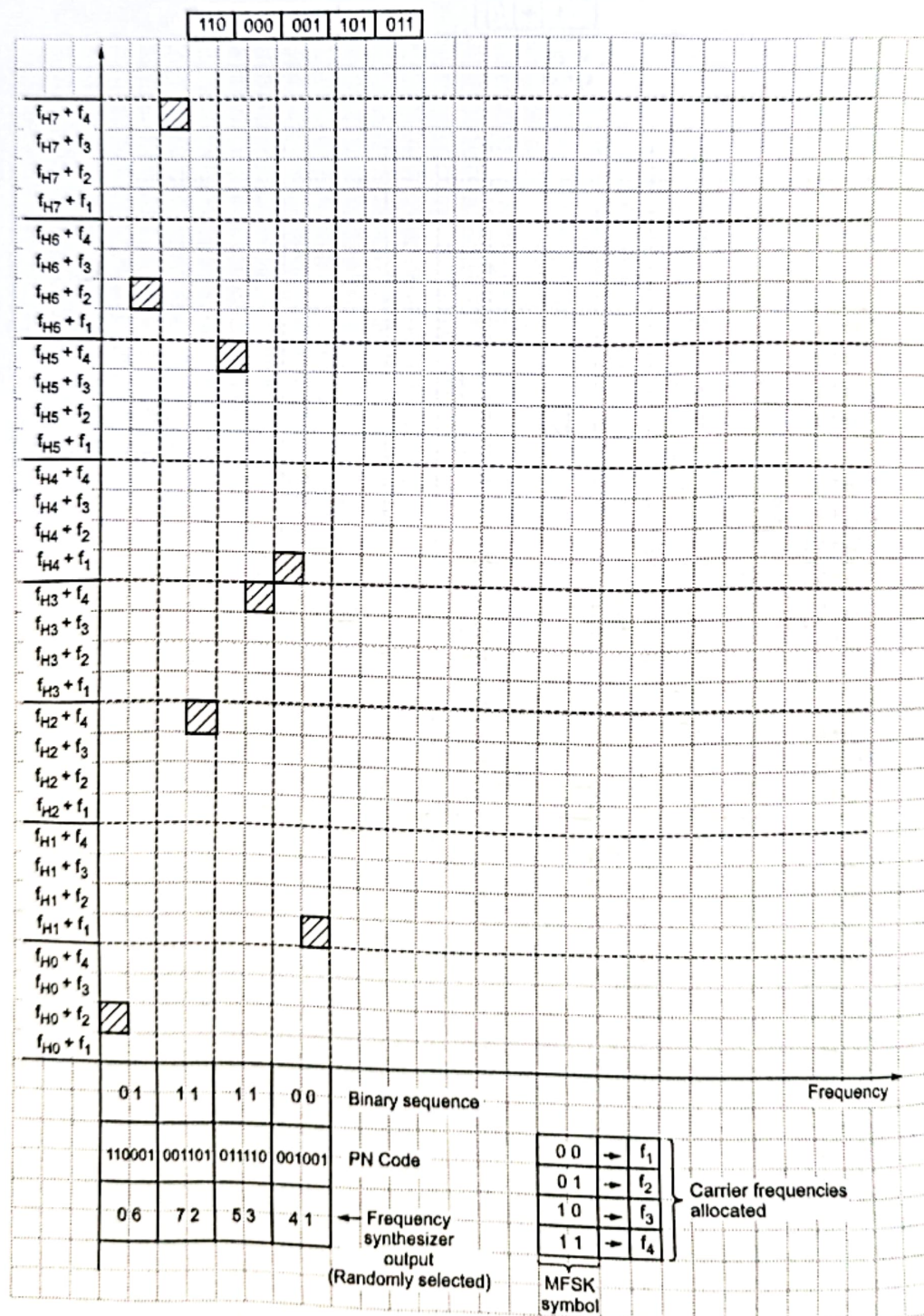
(1F22)Fig. P. 4.10.5

UEx. 4.10.6 SPPU - (E & TC-Sem. 5) May 18, 8 Marks

Represent variation of frequency of fast hop FHSS with binary FSK having following parameters.

- Number of bits per MFSK symbol $K = 2$.
- Number of MFSK tones $M = 2^K = 4$.
- Length of PN segment per hop = 3.
- Total number of frequency hops = 8. Generate PN sequence with initial shift register contents 1100. Represent variation of frequency for binary data 01111100.

✓ Soln. :



(IF25)Fig. P. 4.10.6

4.11 COMPARISON BETWEEN TYPES OF SPREAD SPECTRUM

4.11.1 Comparison of DSSS and FHSS

UQ. Compare DSSS and FHSS.

SPPU - (E&TC-Sem. 5) Dec. 15, 5 Marks, May 16, 5 Marks, May 18, 6 Marks

✓ Ans. :

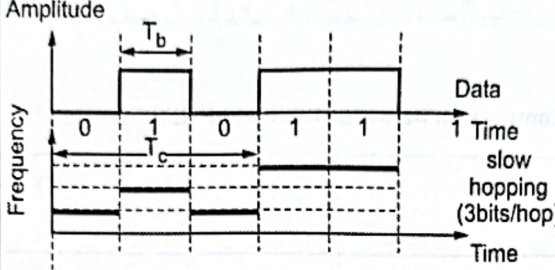
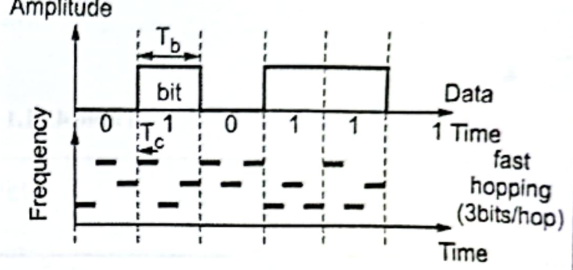
Table 4.11.1 : Comparison between DSSS and FHSS

Sr. No.	Parameter	DSSS	FHSS
1	Abbreviation	Direct sequence spread spectrum.	Frequency hopping spread spectrum.
2	Definition	Wideband PN sequence is Directly multiplied with the narrowband data sequence and wideband spreading code is obtained.	Wideband PN code is used to generate hopping pattern. Data bits are transmitted by changing carrier according to hopping pattern.
3	Chip rate	$R_c = 1/T_c$	$R_c = \max(R_b, R_s)$
4	Modulation technique used	BPSK	MFSK
5	Interference immunity	Low	High
6	Receiver complexity	More complex	Less complex
7	Application	CDMA	Bluetooth technology makes use of fast FHSS

4.11.2 Comparison between Slow FHSS and Fast FHSS

Table 4.11.2 : Comparison between slow FHSS and fast FHSS

Sr. no.	Parameter	Slow FHSS	Fast FHSS
1.	Definition	Multiple symbols are transmitted using one frequency hop.	Multiple hops are used to carry single symbol.
2.	Relation between symbol rate and hop rate	Symbol rate $>$ hop rate	Hop rate $>$ symbol rate
3.	Number of carriers required	Less	More
4.	Detection method used	Coherent detection	Non-coherent detection

Sr. no.	Parameter	Slow FHSS	Fast FHSS
5.	Dwell time	More	Less
6.	waveform	 (1F19)Fig. 4.11.1	 (1F20)Fig. 4.11.2
7.	Application	Wireless LAN	Bluetooth

Chapter Ends

□□□